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Drive circuit of switching element and intelligent power module

Abstract

A gate drive circuit, in order to cause an IGBT in a semiconductor elements to switch to the ON state according to the signal level of an input signal, includes a constant current supply unit configured to supply constant current to the gate of the IGBT, a switching signal input terminal to which a switching signal is input, a signal level determination unit configured to determine a signal level of the switching signal, and a drive capability switching unit configured to, by changing a current amount of constant current output from the constant current supply unit, based on a determination result in the signal level determination unit and a signal level of the input signal, switch the drive capability of the constant current supply unit.

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Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
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12021511	12/2023	Fujii	N/A	H03K 17/567
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Patent No.	Application Date	Country	CPC
2018-143017	12/2017	JP	N/A
2016/009582	12/2015	WO	N/A

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

(1) This application claims the priority benefit of Japanese Patent Application No. JP2022-087586 filed on May 30, 2022, the contents of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The present invention relates to a drive circuit of a switching element that drives the switching element and an intelligent power module including the drive circuit.

BACKGROUND ART

(3) PTL 1 discloses a drive circuit of a voltage control-type device in which turn-on capability of the drive circuit can be adjusted in accordance with specifications or the like of at least one of the voltage control-type device. PTL 2 discloses a programmable drive control circuit in which drive capability of a pre-driver to drive a transistor performing switching operation can be adjusted by software.

CITATION LIST

Patent Literature

(4) PTL 1: WO 2016/9582 A PTL 2: JP 2018-143017 A

SUMMARY OF INVENTION

Technical Problem

(5) Intelligent power modules that integrate a semiconductor chip including an insulated gate bipolar transistor (IGBT) for power conversion and a freewheel diode and an IC for drive and protection function into a package have been known. In such an intelligent power module (IPM), drive capability of a drive circuit to drive an IGBT for power conversion is fixed. Thus, when a signal to bring an IGBT into the ON state is input to the IPM, a metal-oxide-semiconductor (MOS) transistor arranged in the IPM to drive the IGBT operates. The operation of the MOS transistor causes current of a constant current amount to be supplied from the MOS transistor the drive capability of which is fixed to the IGBT, and the IGBT switches from the OFF state to the ON state.

(6) As described above, in the IPM, current supplied from the MOS transistor to the IGBT is fixed to a constant current value, that is, drive capability to drive the IGBT is fixed. Thus, a conventional IPM causes a problem to arise in that, when the drive capability of an IGBT arranged in the IPM is excessively high or insufficient for an object to be driven, such as a motor, an IPM is required to be newly developed.

(7) An object of the present invention is to provide a drive circuit of a switching element and an intelligent power module capable of selecting a drive capability to drive the switching element in a simple method.

Solution to Problem

(8) In order to achieve the above-described object, a drive circuit of a switching element according to one aspect of the present invention, in order to cause a switching element to switch to the ON state according to a signal level of an input signal input from the outside, includes a constant current supply unit configured to supply constant current to a control signal input terminal of the switching element, a switching signal input terminal to which a switching signal to switch drive capability of the constant current supply unit causing the switching element to operate is input, a signal level determination unit configured to determine a signal level of the switching signal, and a drive capability switching unit configured to, by changing a current amount of the constant current, based on a determination result in the signal level determination unit and a signal level of the input signal, switch the drive capability.

(9) In addition, in order to achieve the above-described object, an intelligent power module according to another aspect of the present invention includes a switching element and the drive circuit of the switching element according to the above-described one aspect of the present invention.

Advantageous Effects of Invention

(10) One aspect of the present invention allows a drive capability to drive a switching element to be selected in a simple method.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a block diagram illustrating an example of a schematic configuration of a power conversion device including an intelligent power module according to a first embodiment of the present invention.

(2) FIG. 2 is a circuit block diagram illustrating an example of a gate drive circuit serving as a drive circuit of a switching element according to the first embodiment of the present invention.

(3) FIG. 3 is a circuit block diagram illustrating an example of a gate drive circuit serving as a drive circuit of a switching element according to a second embodiment of the present invention.

DESCRIPTION OF EMBODIMENTS

(4) The embodiments of the present invention indicate devices and methods to embody the technical idea of the present invention by way of example, and the technical idea of the present

invention does not limit the materials, shapes, structures, arrangements, and the like of the constituent components to those described below. The technical idea of the present invention can be subjected to a variety of alterations within the technical scope prescribed by the claims described in CLAIMS.

(5) Although the following description of intelligent power modules according to the respective embodiments will be made below using an inverter device as an example, the intelligent power modules according to the respective embodiments are applicable to not only an inverter device but also a converter device and a modular multilevel converter.

First Embodiment

(6) A drive circuit of a switching element and an intelligent power module according to a first embodiment of the present invention will be described using FIGS. 1 and 2. First, an example of an overall configuration of an intelligent power module 2A according to the present embodiment will be described using FIG. 1. FIG. 1 is a block diagram illustrating an example of a schematic configuration of the intelligent power module 2A according to the present embodiment.

(7) (Overall Configuration of Intelligent Power Module)

(8) As illustrated in FIG. 1, a power conversion device 1A that includes the intelligent power module 2A according to the present embodiment is connected to an AC power unit 4. The AC power unit 4 includes, for example, a three-phase AC power supply (not shown) and a rectifier circuit (not shown) configured to full-wave rectify three-phase AC power input from the three-phase AC power supply. The power conversion device 1A includes a smoothing capacitor 5 configured to smooth power rectified by the rectifier circuit. Although illustration is omitted, the rectifier circuit is, for example, configured by connecting six diodes in a full bridge configuration or connecting six switching elements in a full bridge configuration.

(9) A positive electrode-side line Lp is connected to the positive electrode side of the AC power unit 4 (i.e., a positive electrode output terminal of the rectifier circuit), and a negative electrode-side line Ln is connected to the negative electrode side of the AC power unit 4 (i.e., a negative electrode output terminal of the rectifier circuit). The smoothing capacitor 5 is connected between the positive electrode-side line Lp and the negative electrode-side line Ln. The power conversion device 1A includes the intelligent power module 2A configured to function as an inverter device that converts DC voltage applied between the positive electrode-side line Lp and the negative electrode-side line Ln to three-phase (a U phase, a V phase, and a W phase) AC power and a control device 3 configured to control the intelligent power module 2A.

(10) As illustrated in FIG. 1, the intelligent power module 2A has a positive electrode-side power input terminal Tp to which the positive electrode-side line Lp is connected and a negative electrode-side power input terminal Tn that is connected to the negative electrode-side line Ln. The intelligent power module 2A includes a semiconductor element 20u and a semiconductor element 21u that are connected in series between the positive electrode-side power input terminal Tp and the negative electrode-side power input terminal Tn. The intelligent power module 2A includes a semiconductor element 20v and a semiconductor element 21v that are connected in series between the positive electrode-side power input terminal Tp and the negative electrode-side power input terminal Tn. The intelligent power module 2A includes a semiconductor element 20w and a semiconductor element 21w that are connected in series between the positive electrode-side power input terminal Tp and the negative electrode-side power input terminal Tn.

(11) The semiconductor element 20u and the semiconductor element 21u constitute a U-phase output arm 2U. The semiconductor element 20v and the semiconductor element 21v constitute a V-phase output arm 2V. The semiconductor element 20w and the semiconductor element 21w constitute a W-phase output arm 2W. The semiconductor elements 20u, 20v, and 20w are connected to the positive electrode-side line Lp via the positive electrode-side power input terminal Tp and constitute an upper arm portion. The semiconductor elements 21u, 21v, and 21w are connected to the negative electrode-side line Ln via the negative electrode-side power input terminal Tn and

constitute a lower arm portion.

(12) Each of the semiconductor elements **20u**, **20v**, and **20w** includes an IGBT chip **201** and a freewheel diode **202** connected in inverse parallel to an IGBT **201a** that is arranged in the IGBT chip **201**. An IGBT **201a** and a freewheel diode **202** that are arranged in the semiconductor element **20u** may be formed in an identical semiconductor chip. An IGBT **201a** and a freewheel diode **202** that are arranged in the semiconductor element **20v** may be formed in an identical semiconductor chip. An IGBT **201a** and a freewheel diode **202** that are arranged in the semiconductor element **20w** may be formed in an identical semiconductor chip. The IGBTs **201a** and the freewheel diodes **202** that are respectively arranged in the semiconductor elements **20u**, **20v**, and **20w** are formed in semiconductor chips that are different from one another. Although details will be described later, an IGBT chip **201** arranged in each of the semiconductor elements **20u**, **20v**, and **20w** includes a temperature detection element (not shown in FIG. 1) configured to detect temperature of the IGBT **201a**.

(13) The collector C of an IGBT **201a** and the cathode K of a freewheel diode **202** that are arranged in each of the semiconductor elements **20u**, **20v**, and **20w** are connected to each other and connected to the positive electrode-side power input terminal Tp. The emitter E of an IGBT **201a** and the anode A of a freewheel diode **202** that are arranged in each of the semiconductor elements **20u**, **20v**, and **20w** are connected to each other. An IGBT **201a** arranged in each of the semiconductor elements **20u**, **20v**, and **20w** has a current detection terminal Sn for detecting current that flows through the IGBT **201a** (details will be described later).

(14) Each of the semiconductor elements **21u**, **21v**, and **21w** includes an IGBT chip **211** and a freewheel diode **212** connected in inverse parallel to an IGBT **211a** that is arranged in the IGBT chip **211**. An IGBT **211a** and a freewheel diode **212** that are arranged in the semiconductor element **21u** may be formed in an identical semiconductor chip. An IGBT **211a** and a freewheel diode **212** that are arranged in the semiconductor element **21v** may be formed in an identical semiconductor chip. An IGBT **211a** and a freewheel diode **212** that are arranged in the semiconductor element **21w** may be formed in an identical semiconductor chip. The IGBTs **211a** and the freewheel diodes **212** that are respectively arranged in the semiconductor elements **21u**, **21v**, and **21w** are formed in semiconductor chips that are different from one another. Although details will be described later, an IGBT chip **211** arranged in each of the semiconductor elements **21u**, **21v**, and **21w** includes a temperature detection element (not shown in FIG. 1) configured to detect temperature of the IGBT **211a**.

(15) The collector C of an IGBT **211a** and the cathode K of a freewheel diode **212** that are arranged in each of the semiconductor elements **21u**, **21v**, and **21w** are connected to each other. The collector C of the IGBT **211a** and the cathode K of the freewheel diode **212** that are arranged in the semiconductor element **21u** are connected to the emitter E of the IGBT **201a** and the anode A of the freewheel diode **202** that are arranged in the semiconductor element **20u**. The collector C of the IGBT **211a** and the cathode K of the freewheel diode **212** that are arranged in the semiconductor element **21v** are connected to the emitter E of the IGBT **201a** and the anode A of the freewheel diode **202** that are arranged in the semiconductor element **20v**. The collector C of the IGBT **211a** and the cathode K of the freewheel diode **212** that are arranged in the semiconductor element **21w** are connected to the emitter E of the IGBT **201a** and the anode A of the freewheel diode **202** that are arranged in the semiconductor element **20w**. The emitter E of an IGBT **211a** and the anode A of a freewheel diode **212** that are arranged in each of the semiconductor elements **21u**, **21v**, and **21w** are connected to each other and connected to the negative electrode-side power input terminal Tn. An IGBT **211a** arranged in each of the semiconductor elements **21u**, **21v**, and **21w** has a current detection terminal Sn for detecting current that flows through the IGBT **211a** (details will be described later).

(16) The emitter E of the IGBT and the anode A of the freewheel diode **202** that are arranged in the semiconductor element **20u** and the collector C of the IGBT and the cathode K of the freewheel

diode **212** that are arranged in the semiconductor element **21u** are connected to a U-phase output terminal TU. The U-phase output terminal TU is a terminal to which U-phase AC voltage that the intelligent power module **2A** generated by DC-AC converting the DC voltage input from the AC power unit **4** is output.

(17) The emitter E of the IGBT and the anode A of the freewheel diode **202** that are arranged in the semiconductor element **20v** and the collector C of the IGBT and the cathode K of the freewheel diode **212** that are arranged in the semiconductor element **21v** are connected to a V-phase output terminal TV. The V-phase output terminal TV is a terminal to which V-phase AC voltage that the intelligent power module **2A** generated by DC-AC converting the DC voltage input from the AC power unit **4** is output.

(18) The emitter E of the IGBT and the anode A of the freewheel diode **202** that are arranged in the semiconductor element **20w** and the collector C of the IGBT and the cathode K of the freewheel diode **212** that are arranged in the semiconductor element **21w** are connected to a W-phase output terminal TW. The W-phase output terminal TW is a terminal to which W-phase AC voltage that the intelligent power module **2A** generated by DC-AC converting the DC voltage input from the AC power unit **4** is output.

(19) As illustrated in FIG. **1**, for example, a motor **7** that serves as a load is connected to the U-phase output terminal TU, the V-phase output terminal TV, and the W-phase output terminal TW that are arranged in the intelligent power module **2A**. Thus, the emitters E of the IGBTs **201a** and the anodes A of the freewheel diodes **202** that are respectively arranged in the semiconductor elements **20u**, **20v**, and **20w** and the collectors C of the IGBTs **211a** and the cathodes K of the freewheel diodes **212** that are respectively arranged in the semiconductor elements **21u**, **21v**, and **21w** are connected to the motor **7**.

(20) As illustrated in FIG. **1**, the intelligent power module **2A** includes a gate drive circuit **22A** configured to drive the semiconductor elements **20u**, **20v**, **20w**, **21u**, **21v**, and **21w**. Each of the IGBTs **201a** that are respectively arranged in the semiconductor elements **20u**, **20v**, and **20w** and the IGBTs **211a** that are respectively arranged in the semiconductor elements **21u**, **21v**, and **21w** is equivalent to an example of the switching element. In addition, the gate drive circuit **22A** is equivalent to an example of the drive circuit of the switching element according to the present embodiment. Therefore, the intelligent power module **2A** includes the IGBTs (an example of the switching element) **201a** and **211a** and the gate drive circuit **22A** (an example of the drive circuit of the switching element according to the present embodiment).

(21) As illustrated in FIG. **1**, in order to cause the IGBT **201a** of the semiconductor element **20u** to switch to the ON state according to a signal level of an input signal SinUH input from the control device (an example of the outside) **3** (details will be described later), the gate drive circuit **22A** includes a constant current supply unit **221uA** configured to supply constant current to the gate (an example of a control signal input terminal) G of the IGBT **201a**. The gate drive circuit **22A** has a signal input terminal Tiuh to which the input signal SinUH that the control device **3** outputs is input. The constant current supply unit **221uA** is connected to the signal input terminal Tiuh. This configuration allows the input signal SinUH to be input to the constant current supply unit **221uA**.

(22) The gate drive circuit **22A** has a switching signal input terminal Tsw1 to which a switching signal Ssw1 to switch drive capability of the constant current supply unit **221uA**, which drives the IGBT **201a** arranged in the semiconductor element **20u**, is input. To the switching signal input terminal Tsw1, a Zener diode **6uH** that generates the switching signal Ssw1 and also determines a signal level of the switching signal Ssw1 is connected. The Zener diode **6uH** is connected between the switching signal input terminal Tsw1 and a reference potential terminal T22 (for example, a ground terminal). The cathode K of the Zener diode **6uH** is connected to the switching signal input terminal Tsw1, and the anode A of the Zener diode **6uH** is connected to the reference potential terminal T22.

(23) The gate drive circuit **22A** includes a signal level determination unit **227uA** configured to

determine a signal level of the switching signal Ssw1. The gate drive circuit 22A includes a drive capability switching unit 222uA configured to change a current amount of constant current output from the constant current supply unit 221uA, based on a determination result in the signal level determination unit 227uA and a signal level of the input signal SinUH and thereby switch the drive capability of the constant current supply unit 221uA. An output terminal of the signal level determination unit 227uA is connected to an input terminal of the drive capability switching unit 222uA. By way of this, an output signal including information about a determination result in which a signal level of the switching signal Ssw1 input from the switching signal input terminal Tsw1 is determined in the signal level determination unit 227uA can be input to the drive capability switching unit 222uA.

(24) In order to cause the IGBT 201a of the semiconductor element 20v to switch to the ON state according to a signal level of an input signal SinVH input from the control device 3, the gate drive circuit 22A includes a constant current supply unit 221vA configured to supply constant current to the gate G of the IGBT 201a. The gate drive circuit 22A has a signal input terminal Tivh to which the input signal SinVH that the control device 3 outputs is input. The constant current supply unit 221vA is connected to the signal input terminal Tivh. This configuration allows the input signal SinVH to be input to the constant current supply unit 221vA.

(25) The gate drive circuit 22A has a switching signal input terminal Tsw2 to which a switching signal Ssw2 to switch drive capability of the constant current supply unit 221vA, which drives the IGBT 201a of the semiconductor element 20v, is input. To the switching signal input terminal Tsw2, a Zener diode 6vH that generates the switching signal Ssw2 and also determines a signal level of the switching signal Ssw2 is connected. The Zener diode 6vH is connected between the switching signal input terminal Tsw2 and the reference potential terminal T22. The cathode K of the Zener diode 6vH is connected to the switching signal input terminal Tsw2, and the anode A of the Zener diode 6vH is connected to the reference potential terminal T22.

(26) The gate drive circuit 22A includes a signal level determination unit 227vA configured to determine a signal level of the switching signal Ssw2. The gate drive circuit 22A includes a drive capability switching unit 222vA configured to change a current amount of constant current output from the constant current supply unit 221vA, based on a determination result in the signal level determination unit 227vA and a signal level of the input signal SinVH and thereby switch the drive capability of the constant current supply unit 221vA. An output terminal of the signal level determination unit 227vA is connected to an input terminal of the drive capability switching unit 222vA. By way of this, an output signal including information about a determination result output from the signal level determination unit 227vA can be input to the drive capability switching unit 222vA.

(27) In order to cause the IGBT 201a arranged in the semiconductor element 20w to switch to the ON state according to a signal level of an input signal SinWH input from the control device 3, the gate drive circuit 22A includes a constant current supply unit 221wA configured to supply constant current to the gate G formed in the IGBT 201a. The gate drive circuit 22A has a signal input terminal Tiwh to which the input signal SinWH that the control device 3 outputs is input. The constant current supply unit 221wA is connected to the signal input terminal Tiwh. This allows the input signal SinWH to be input to the constant current supply unit 221wA.

(28) The gate drive circuit 22A has a switching signal input terminal Tsw3 to which a switching signal Ssw3 to switch drive capability of the constant current supply unit 221wA, which drives the IGBT 201a of the semiconductor element 20w, is input. To the switching signal input terminal Tsw3, a Zener diode 6wH that generates the switching signal Ssw3 and also determines a signal level of the switching signal Ssw3 is connected. The Zener diode 6wH is connected between the switching signal input terminal Tsw3 and the reference potential terminal T22. The cathode K of the Zener diode 6wH is connected to the switching signal input terminal Tsw3, and the anode A of the Zener diode 6wH is connected to the reference potential terminal T22.

(29) The gate drive circuit **22A** includes a signal level determination unit **227wA** configured to determine a signal level of the switching signal **Ssw3**. The gate drive circuit **22A** includes a drive capability switching unit **222wA** configured to change a current amount of constant current output from the constant current supply unit **221wA**, based on a determination result in the signal level determination unit **227wA** and a signal level of the input signal **SinWH** and thereby switch the drive capability of the constant current supply unit **221wA**. An output terminal of the signal level determination unit **227wA** is connected to an input terminal of the drive capability switching unit **222wA**. By way of this, an output signal including information about a determination result output from the signal level determination unit **227wA** can be input to the drive capability switching unit **222wA**.

(30) In order to cause the IGBT **211a** arranged in the semiconductor element **21u** to switch to the ON state according to a signal level of an input signal **SinUL** input from the control device **3**, the gate drive circuit **22A** includes a constant current supply unit **223uA** configured to supply constant current to the gate **G** formed in the IGBT **211a**. The gate drive circuit **22A** has a signal input terminal **Tiul** to which the input signal **SinUL** that the control device **3** outputs is input. The constant current supply unit **223uA** is connected to the signal input terminal **Tiul**. This allow the input signal **SinUL** to be input to the constant current supply unit **223uA**.

(31) The gate drive circuit **22A** has a switching signal input terminal **Tsw4** to which a switching signal **Ssw4** to switch drive capability of the constant current supply unit **223uA**, which drives the IGBT **211a** of the semiconductor element **21u**, is input. To the switching signal input terminal **Tsw4**, a Zener diode **6uL** that generates the switching signal **Ssw4** and also determines a signal level of the switching signal **Ssw4** is connected. The Zener diode **6uL** is connected between the switching signal input terminal **Tsw4** and the reference potential terminal **T22**. The cathode **K** of the Zener diode **6uL** is connected to the switching signal input terminal **Tsw4**, and the anode **A** of the Zener diode **6uL** is connected to the reference potential terminal **T22**.

(32) The gate drive circuit **22A** includes a signal level determination unit **226uA** configured to determine a signal level of the switching signal **Ssw4**. The gate drive circuit **22A** includes a drive capability switching unit **224uA** configured to change a current amount of constant current output from the constant current supply unit **223uA**, based on a determination result in the signal level determination unit **226uA** and a signal level of the input signal **SinUL** and thereby switch the drive capability of the constant current supply unit **223uA**. An output terminal of the signal level determination unit **226uA** is connected to an input terminal of the drive capability switching unit **224uA**. By way of this, an output signal including information about a determination result output from the signal level determination unit **226uA** can be input to the drive capability switching unit **224uA**.

(33) In order to cause the IGBT **211a** arranged in the semiconductor element **21v** to switch to the ON state according to a signal level of an input signal **SinVL** input from the control device **3**, the gate drive circuit **22A** includes a constant current supply unit **223vA** configured to supply constant current to the gate **G** of the IGBT **211a**. The gate drive circuit **22A** has a signal input terminal **Tivl** to which the input signal **SinVL** that the control device **3** outputs is input. The constant current supply unit **223vA** is connected to the signal input terminal **Tivl**. This allows the input signal **SinVL** to be input to the constant current supply unit **223vA**.

(34) The gate drive circuit **22A** has a switching signal input terminal **Tsw5** to which a switching signal **Ssw5** to switch drive capability of the constant current supply unit **223vA**, which drives the IGBT **211a** of the semiconductor element **21v**, is input. To the switching signal input terminal **Tsw5**, a Zener diode **6vL** that generates the switching signal **Ssw5** and also determines a signal level of the switching signal **Ssw5** is connected. The Zener diode **6vL** is connected between the switching signal input terminal **Tsw5** and the reference potential terminal **T22**. The cathode **K** of the Zener diode **6vL** is connected to the switching signal input terminal **Tsw5**, and the anode **A** of the Zener diode **6vL** is connected to the reference potential terminal **T22**.

(35) The gate drive circuit **22A** includes a signal level determination unit **226vA** configured to determine a signal level of the switching signal **Ssw5**. The gate drive circuit **22A** includes a drive capability switching unit **224vA** configured to change a current amount of constant current output from the constant current supply unit **223vA**, based on a determination result in the signal level determination unit **226vA** and a signal level of the input signal **SinVL** and thereby switch the drive capability of the constant current supply unit **223vA**. An output terminal of the signal level determination unit **226vA** is connected to an input terminal of the drive capability switching unit **224vA**. By way of this, an output signal including information about a determination result output from the signal level determination unit **226vA** can be input to the drive capability switching unit **224vA**.

(36) In order to cause the IGBT **211a** arranged in the semiconductor element **21w** to switch to the ON state according to a signal level of an input signal **SinWL** input from the control device **3**, the gate drive circuit **22A** includes a constant current supply unit **223wA** configured to supply constant current to the gate **G** of the IGBT **211a**. The gate drive circuit **22A** has a signal input terminal **Tiwl** to which the input signal **SinWL** that the control device **3** outputs is input. The constant current supply unit **223wA** is connected to the signal input terminal **Tiwl**. This allows the input signal **SinWL** to be input to the constant current supply unit **223wA**.

(37) The gate drive circuit **22A** has a switching signal input terminal **Tsw6** to which a switching signal **Ssw6** to switch drive capability of the constant current supply unit **223wA**, which drives the IGBT **211a** of the semiconductor element **21w**, is input. To the switching signal input terminal **Tsw6**, a Zener diode **6wL** that generates the switching signal **Ssw6** and also determines a signal level of the switching signal **Ssw6** is connected. The Zener diode **6wL** is connected between the switching signal input terminal **Tsw6** and the reference potential terminal **T22**. The cathode **K** of the Zener diode **6wL** is connected to the switching signal input terminal **Tsw6**, and the anode **A** of the Zener diode **6wL** is connected to the reference potential terminal **T22**.

(38) The gate drive circuit **22A** includes a signal level determination unit **226wA** configured to determine a signal level of the switching signal **Ssw6**. The gate drive circuit **22A** includes a drive capability switching unit **224wA** configured to change a current amount of constant current output from the constant current supply unit **223wA**, based on a determination result in the signal level determination unit **226wA** and a signal level of the input signal **SinWL** and thereby switch the drive capability of the constant current supply unit **223wA**. An output terminal of the signal level determination unit **226wA** is connected to an input terminal of the drive capability switching unit **224wA**. By way of this, an output signal including information about a determination result output from the signal level determination unit **226wA** can be input to the drive capability switching unit **224wA**.

(39) Although illustration is omitted, the gate drive circuit **22A** has a power source terminal to which control voltage to control the constant current supply unit **221uA** is input, a power source terminal to which control voltage to control the constant current supply unit **221vA** is input, a power source terminal to which control voltage to control the constant current supply unit **221wA** is input, and power source terminals to which control voltages to control the constant current supply unit **223uA**, the constant current supply unit **223vA**, and the constant current supply unit **223wA** are input.

(40) Hereinafter, the semiconductor elements **20u**, **20v**, **20w**, **21u**, **21v**, and **21w** may simply be referred to as “semiconductor elements **20u** to **21w**”. The constant current supply units **221uA**, **221vA**, **221wA**, **223uA**, **223vA**, and **223wA** may simply be referred to as “constant current supply units **221uA** to **223wA**”. Further, the drive capability switching units **222uA**, **222vA**, **222wA**, **224uA**, **224vA**, and **224wA** may simply be referred to as “drive capability switching units **222uA** to **224wA**”. Further, the input signals **SinUH**, **SinVH**, **SinWH**, **SinUL**, **SinVL**, and **SinWL** may simply be referred to as “input signals **SinUH** to **SinWL**”. Further, the Zener diodes **6uH**, **6vH**, **6wH**, **6uL**, **6vL**, and **6wL** may simply be referred to as “Zener diodes **6uH** to **6wL**”. Further, the switching

signal input terminals Tsw1, Tsw2, Tsw3, Tsw4, Tsw5, and Tsw6 may simply be referred to as “switching signal input terminals Tsw1 to Tsw6”. Further, alarm signal output terminals VFOu, VFOv, VFOw, and VFO may simply be referred to as “alarm signal output terminals VFOu to VFO”.

(41) As illustrated in FIG. 1, the gate drive circuit 22A includes a protection unit 225uA configured to protect the IGBT 201a arranged in the semiconductor element 20u from overcurrent or a high-temperature condition and protect the constant current supply unit 221uA from a malfunction caused by reduction in control voltage. The gate drive circuit 22A includes a protection unit 225vA configured to protect the IGBT 201a arranged in the semiconductor element 20v from overcurrent or a high-temperature condition and protect the constant current supply unit 221vA from a malfunction caused by reduction in control voltage. The gate drive circuit 22A includes a protection unit 225wA configured to protect the IGBT 201a arranged in the semiconductor element 20w from overcurrent or a high-temperature condition and protect the constant current supply unit 221wA from a malfunction caused by reduction in control voltage. The gate drive circuit 22A includes a protection unit 225 configured to protect the IGBTs 211a respectively arranged in the semiconductor elements 21u, 21v, and 21w from overcurrent or a high-temperature condition and protect the constant current supply units 223uA, 223vA, and 223wA from a malfunction caused by reduction in control voltage. As described above, the gate drive circuit 22A includes the protection units 225uA, 225vA, and 225wA for the U-phase, the V-phase, and the W-phase, respectively, on the upper arm side and includes the protection unit 225 shared by the U-phase, the V-phase, and the W-phase on the lower arm side. Details of the protection units 225uA, 225vA, 225wA, and 225 will be described later.

(42) As illustrated in FIG. 1, the gate drive circuit 22A has an alarm signal output terminal VFOu that is connected to the protection unit 225uA and to which an alarm signal ALMu generated by the protection unit 225uA is output. The gate drive circuit 22A has an alarm signal output terminal VFOv that is connected to the protection unit 225vA and to which an alarm signal ALMv generated by the protection unit 225vA is output. The gate drive circuit 22A has an alarm signal output terminal VFOw that is connected to the protection unit 225wA and to which an alarm signal ALMw generated by the protection unit 225wA is output. The gate drive circuit 22A has an alarm signal output terminal VFO that is connected to the protection unit 225 and to which an alarm signal ALM generated by the protection unit 225 is output.

(43) The drive capability of the gate drive circuit 22A is configured to be able to be switched with respect to each of the U-phase, the V-phase, and the W-phase in each of the upper arm and the lower arm by Zener voltages of the Zener diodes 6uH to 6wL. Further, the intelligent power module 2A is configured such that a user of the intelligent power module 2A can mount the Zener diodes 6uH to 6wL to the switching signal input terminals Tsw1 to Tsw6, respectively. Thus, in the gate drive circuit 22A and the intelligent power module 2A, the drive capability can be switched by a simple method of mounting the Zener diodes 6uH to 6wL to the switching signal input terminals Tsw1 to Tsw6, respectively.

(44) Although details will be described later, the alarm signal output terminals VFOu to VFO have an open-drain configuration. Thus, when the protection units 225uA, 225vA, 225wA, and 225 do not detect an abnormality of the IGBTs 201a and 211a and the like, the voltage levels of the alarm signal output terminals VFOu to VFO are a high level, respectively. In contrast, when the protection units 225uA, 225vA, 225wA, and 225 detect an abnormality of the IGBTs 201a and 211a and the like, the alarm signals ALMu, ALMv, ALMw, and ALM the voltage levels (i.e., signal levels) of which are a low level (for example, the voltage level of the reference potential terminal (for example, 0 V)) are output, respectively.

(45) As illustrated in FIG. 1, the control device 3 is configured to output the input signals SinUH to Sin WL having, for example, a pulse shape to the gate drive circuit 22A. By way of this, the control device 3 is capable of, by controlling the gate drive circuit 22A, controlling the IGBTs 201a

included in the respective ones of the semiconductor elements **20u**, **20v**, and **20w** and the IGBTs **211a** included in the respective ones of the semiconductor elements **21u**, **21v**, and **21w** by, for example, pulse width modulation (PWM).

(46) (Configuration of Drive Circuit of Switching Element)

(47) Next, a schematic configuration of the gate drive circuit **22A** serving as the drive circuit of switching elements according to the present embodiment will be described using FIG. **2** with reference to FIG. **1**. The constant current supply units **221uA** to **223wA** have the same configuration as one another. Thus, the configurations of the constant current supply units **221uA** to **223wA** will be described using the constant current supply unit **223uA** as an example. The drive capability switching units **222uA** to **224wA** have the same configuration as one another. Thus, the configurations of the drive capability switching units **222uA** to **224wA** will be described using the drive capability switching unit **224uA** as an example.

(48) FIG. **2** is a circuit block diagram illustrating a schematic configuration of the gate drive circuit **22A**. In FIG. **2**, among the constituent components arranged in the gate drive circuit **22A**, the constant current supply unit **223uA**, the drive capability switching unit **224uA**, and the protection unit **225** are illustrated. Further, in FIG. **2**, to facilitate understanding, the semiconductor element **21u** including the IGBT **211a** to which the constant current supply unit **223uA** supplies current, the Zener diode **6uL** connected to the switching signal input terminal **Tsw4**, and a power source **8L** connected to a power source terminal **Tvin** are also illustrated.

(49) As illustrated in FIG. **2**, the constant current supply unit **223uA** includes a constant current source **223e** connected to the power source terminal **Tvin** and a resistance element **223c** and a resistance element **223d** connected in series between the constant current source **223e** and a reference potential terminal **T223**. One terminal of the resistance element **223c** is connected to a current output terminal of the constant current source **223e**, and the other terminal of the resistance element **223c** is connected to one terminal of the resistance element **223d**. A connection point between the output terminal of the constant current source **223e** and the one terminal of the resistance element **223c** is connected to the signal input terminal **Tiul**.

(50) The other terminal of the resistance element **223d** is connected to the reference potential terminal **T223**. The constant current supply unit **223uA** is a constituent element that drives the semiconductor element **21u** constituting the lower arm. Thus, the reference potential terminal **T223** is connected to the reference potential terminal **T22** and, for example, connected to the ground terminal. Note that reference potential terminals (not shown) that are arranged in the constant current supply units **221uA**, **221vA**, and **221wA** (see FIG. **1**) driving the semiconductor elements **20u**, **20v**, and **20w** constituting the upper arm and correspond to the reference potential terminal **T223** are, for example, connected to the emitters **E** of the IGBTs **201a** (see FIG. **1**) arranged in the semiconductor elements **20u**, **20v**, and **20w**.

(51) The constant current supply unit **223uA** includes a comparator **223a** formed by, for example, an operational amplifier and a voltage generation unit **223b** connected to the non-inverting input terminal (+) of the comparator **223a**. The inverting input terminal (−) of the comparator **223a** is connected to a connection point between the other terminal of the resistance element **223c** and the one terminal of the resistance element **223d**. The output terminal of the comparator **223a** is connected to the drive capability switching unit **224uA**.

(52) The voltage generation unit **223b** is configured by, for example, a constant voltage source. The negative electrode side of the voltage generation unit **223b** is connected to the reference potential terminal **T223**. The positive electrode side of the voltage generation unit **223b** is connected to the non-inverting input terminal (+) of the comparator **223a**. The voltage generation unit **223b** is configured to generate a comparison voltage **V0** that serves as a reference for comparison in the comparator **223a**. The comparison voltage **V0** is set to a voltage level that allows which one of the high level and the low level the signal level of the input signal **SinUL** input from the signal input terminal **Tiul** is to be detected. To the inverting input terminal (−) of the comparator **223a**, a

divided voltage obtained by resistance-dividing the signal level (i.e., voltage level) of the input signal SinUL by the resistance value of the resistance element **223c** and the resistance value of the resistance element **223d** is input. Thus, the voltage level of the comparison voltage **V0** is set to be a voltage between a divided voltage in the case where the signal level of the input signal SinUL is a maximum level and a divided voltage in the case where the signal level of the input signal SinUL is a minimum level.

(53) By way this, the comparator **223a** is capable of determining whether the signal level of the input signal SinUL input from the signal input terminal Tiul is the high level or the low level. When the voltage level of the divided voltage input to the inverting input terminal (−) is lower than the voltage level of the comparison voltage **V0**, that is, when the signal level of the input signal SinUL is the low level, the comparator **223a** outputs an output signal So0 the signal level of which is the high level to the drive capability switching unit **224uA**. On the other hand, when the voltage level of the divided voltage input to the inverting input terminal (−) is higher than the voltage level of the comparison voltage **V0**, that is, when the signal level of the input signal SinUL is the high level, the comparator **223a** outputs the output signal So0 the signal level of which is the low level to the drive capability switching unit **224uA**.

(54) As illustrated in FIG. 2, the constant current supply unit **223uA** includes a plurality of (two in the present embodiment) transistors **223f** and **223g** configured to supply constant current to the gate G of the IGBT **211a** arranged in the semiconductor element **21u**. The transistor **223f** and the transistor **223g** are respectively formed by, for example, P-type MOS field effect transistors (FETs). The source S of each of the transistor **223f** and the transistor **223g** is connected to the power source terminal Tvin. The drain D of each of the transistor **223f** and the transistor **223g** is connected to the gate G of the IGBT **211a** arranged in the semiconductor element **21u**. The gate G of the transistor **223f** and the gate G of the transistor **223g** are respectively connected to the drive capability switching unit **224uA**.

(55) Current amounts of constant currents that the transistors **223f** and **223g** (an example of at least two transistors among a plurality of transistors) respectively supplies to the gate G of the IGBT **211a** arranged in the semiconductor element **21u** are different from each other. For example, the transistor **223f** has a larger transistor size than the transistor **223g**. By way of this, the current amount of the constant current from the transistor **223f** is larger than the current amount of the constant current from the transistor **223g**.

(56) Although details will be described later, in the case where the transistor **223f** is selected as a transistor to supply constant current to the IGBT **211a** arranged in the semiconductor element **21u**, the constant current supply unit **223uA** is capable of causing the IGBT **211a** to switch to the ON state with higher drive capability than a case where the transistor **223g** is selected. On the other hand, in the case where the transistor **223g** is selected as a transistor to supply constant current to the IGBT **211a** arranged in the semiconductor element **21u**, the constant current supply unit **223uA** is capable of causing the IGBT **211a** to switch to the ON state with lower drive capability than a case where the transistor **223f** is selected. As described above, the constant current supply unit **223uA** is capable of, by changing a transistor to be caused to operate between the transistor **223g** and the transistor **223f**, switching drive capability to drive the IGBT **211a**.

(57) As illustrated in FIG. 2, the protection unit **225** included in the gate drive circuit **22A** includes a temperature detection circuit (an example of a temperature detection unit) **225f** configured to detect temperature of the IGBT **211a**, a current detection circuit (an example of a current detection unit) **225g** configured to detect current flowing through the IGBT **211a**, a control voltage detection circuit (an example of a control voltage detection unit) **225e** configured to detect control voltage to control the constant current supply unit **223uA**, and an alarm signal generation circuit (an example of an alarm signal generation unit) **225d** configured to generate an alarm signal ALM notifying that at least one of the temperature detection circuit **225f**, the current detection circuit **225g**, and the control voltage detection circuit **225e** has detected an abnormality.

(58) As illustrated in FIG. 2, a temperature detection element **211b** arranged in the IGBT chip **211** is formed by, for example, a diode. The diode is formed of, for example, silicon. The temperature detection element **211b** is connected to the temperature detection circuit **225f**. The temperature detection circuit **225f** includes a comparator **225f-1** configured to compare a comparison signal set to a predetermined signal level with a detection signal detected by the temperature detection element **211b**. The temperature detection circuit **225f** includes a voltage generation unit **225f-2** configured to generate a voltage of the comparison signal. The comparator **225f-1** is formed by, for example, an operational amplifier. The voltage generation unit **225f-2** is formed by, for example, a constant voltage source. The voltage generation unit **225f-2** generates, for example, a voltage corresponding to the absolute maximum rated temperature of the IGBT **211a**.

(59) The anode of the temperature detection element **211b** is connected to the inverting input terminal (−) of the comparator **225f-1**, and the cathode of the temperature detection element **211b** is connected to the reference potential terminal T22. The non-inverting input terminal (+) of the comparator **225f-1** is connected to the positive electrode side of the voltage generation unit **225f-2**. The output terminal of the comparator **225f-1** is connected to an input terminal of the alarm signal generation circuit **225d**. The negative electrode side of the voltage generation unit **225f-2** is connected to the reference potential terminal T22.

(60) Although illustration is omitted, to the anode of the temperature detection element **211b**, a constant current source is connected. The forward voltage of a diode made of silicon is generally lower in the case where ambient temperature is high than in the case where the ambient temperature is low. Thus, when the temperature of the IGBT **211a** increases in the case where constant current is input from the constant current source to the temperature detection element **211b**, the voltage drop in the temperature detection element **211b** becomes small. As a result, the voltage input from the temperature detection element **211b** to the temperature detection circuit **225f** decreases as the temperature of the IGBT **211a** increases.

(61) Thus, when the temperature of the IGBT **211a** is lower than the absolute maximum rated temperature, since voltage detected by the temperature detection element **211b** is higher than the voltage of the comparison signal generated by the voltage generation unit **225f-2**, the comparator **225f-1** outputs an output signal the signal level of which is the low level to the alarm signal generation circuit **225d**. In contrast, when the temperature of the IGBT **211a** is higher than the absolute maximum rated temperature, since voltage detected by the temperature detection element **211b** is lower than the voltage of the comparison signal generated by the voltage generation unit **225f-2**, the comparator **225f-1** outputs an output signal the signal level of which is the high level to the alarm signal generation circuit **225d**. As described above, the temperature detection circuit **225f** is capable of, by using voltage that is input from the temperature detection element **211b** and changes according to the temperature of the IGBT **211a**, detecting whether or not the temperature of the IGBT **211a** exceeds the absolute maximum rated temperature of the IGBT **211a**.

(62) As illustrated in FIG. 2, the current detection circuit **225g** includes a resistance element **225g-3** connected between the current detection terminal Sn of the IGBT **211a** and the reference potential terminal T22. One terminal of the resistance element **225g-3** is connected to the current detection terminal Sn, and the other terminal of the resistance element **225g-3** is connected to the reference potential terminal T22. The current detection circuit **225g** is configured to output, as detection voltage, a voltage drop occurring across the resistance element **225g-3** caused by detection current that is output from the current detection terminal Sn flowing through the resistance element **225g-3**. In other words, the current detection circuit **225g** outputs voltage at one terminal of the resistance element **225g-3** connected to the current detection terminal Sn as detection voltage.

(63) The current detection circuit **225g** includes a comparator **225g-1** connected to the one terminal of the resistance element **225g-3** and a voltage generation unit **225g-2** configured to generate reference voltage serving as a reference for comparison in the comparator **225g-1**.

(64) The voltage generation unit **225g-2** is formed by, for example, a DC power source. The

negative electrode side of the voltage generation unit **225g-2** is connected to the reference potential terminal **T22**. The voltage generation unit **225g-2** is configured to generate a comparison voltage having a predetermined voltage level. The comparison voltage is set to, for example, a voltage corresponding to the absolute maximum rated current of the IGBT **211a**.

(65) The comparator **225g-1** is formed by, for example, an operational amplifier. To the inverting input terminal (−) of the comparator **225g-1**, the current detection terminal **Sn** and one terminal of the resistance element **225g-3** are connected. To the non-inverting input terminal (+) of the comparator **225g-1**, the positive electrode side of the voltage generation unit **225g-2** is connected. The output terminal of the comparator **225g-1** is connected to an input terminal of the alarm signal generation circuit **225d**. When detection voltage that is a voltage drop across the resistance element **225g-3** is lower than the comparison voltage generated by the voltage generation unit **225g-2**, the comparator **225g-1** outputs an output signal the signal level (i.e., a voltage level) of which is the low level to the alarm signal generation circuit **225d**. In addition, when the detection voltage is higher than the comparison voltage, the comparator **225g-1** outputs an output signal the signal level of which is the high level to the alarm signal generation circuit **225d**.

(66) The detection current output from the current detection terminal **Sn** is proportional to current output from the emitter **E** of the IGBT **211a**. Thus, when voltage corresponding to the detection current output from the current detection terminal **Sn** is lower than voltage corresponding to the absolute maximum rated current of the IGBT **211a** (i.e., when the IGBT **211a** is in a normal state), the current detection circuit **225g** outputs an output signal the signal level of which is the low level. In contrast, when voltage corresponding to the detection current output from the current detection terminal **Sn** is higher than voltage corresponding to the absolute maximum rated current of the IGBT **211a** (i.e., when overcurrent is flowing through the IGBT **211a**), the current detection circuit **225g** outputs an output signal the signal level of which is the high level.

(67) As illustrated in FIG. 2, the control voltage detection circuit **225e** includes a comparator **225e-1** connected to the power source terminal **Tvin** and a voltage generation unit **225e-2** configured to generate reference voltage serving as a reference for comparison in the comparator **225e-1**.

(68) The voltage generation unit **225e-2** is formed by, for example, a DC power source. The negative electrode side of the voltage generation unit **225e-2** is connected to the reference potential terminal **T22**. The voltage generation unit **225e-2** is configured to generate a comparison voltage having a predetermined voltage level. The comparison voltage is set to, for example, a minimum voltage at which the constant current supply unit **223uA** can operate.

(69) The comparator **225e-1** is formed by, for example, an operational amplifier. To the inverting input terminal (−) of the comparator **225e-1**, the power source terminal **Tvin** is connected. To the non-inverting input terminal (+) of the comparator **225e-1**, the positive electrode side of the voltage generation unit **225e-2** is connected. The output terminal of the comparator **225e-1** is connected to an input terminal of the alarm signal generation circuit **225d**. When the control voltage input from the power source terminal **Tvin** is higher than the comparison voltage generated by the voltage generation unit **225e-2**, the comparator **225e-1** outputs an output signal the signal level (i.e., a voltage level) of which is the low level to the alarm signal generation circuit **225d**. In addition, when the control voltage is lower than the comparison voltage, the comparator **225e-1** outputs an output signal the signal level of which is the high level to the alarm signal generation circuit **225d**.

(70) Thus, when the control voltage input from the power source terminal **Tvin** is higher than the minimum voltage at which the constant current supply unit **223uA** can operate (i.e., when the constant current supply unit **223uA** can normally operate), the control voltage detection circuit **225e** outputs an output signal the signal level of which is the low level. In contrast, when the control voltage input from the power source terminal **Tvin** is lower than the minimum voltage at which the constant current supply unit **223uA** can operate (i.e., when the constant current supply unit **223uA** cannot normally operate), the control voltage detection circuit **225e** outputs an output signal the signal level of which is the high level.

(71) As described above, each of the temperature detection circuit **225f**, the current detection circuit **225g**, and the control voltage detection circuit **225e** outputs an output signal the signal level of which is the low level to the alarm signal generation circuit **225d** when detecting that a detection target is in a normal state and outputs an output signal the signal level of which is the high level to the alarm signal generation circuit **225d** when detecting that the detection target is in an abnormal state.

(72) Although illustration is omitted, the protection unit **225** includes a temperature detection circuit that has the same configuration as the temperature detection circuit **225f** and detects temperature of the IGBT **211a** arranged in the semiconductor element **21v** and a current detection circuit that has the same configuration as the current detection circuit **225g** and detects current flowing through the IGBT **211a** arranged in the semiconductor element **21v**. Further, the protection unit **225** includes a temperature detection circuit that has the same configuration as the temperature detection circuit **225f** and detects temperature of the IGBT **211a** arranged in the semiconductor element **21w** and a current detection circuit that has the same configuration as the current detection circuit **225g** and detects current flowing through the IGBT **211a** arranged in the semiconductor element **21w**. Output signals output from the detection circuits are input to the alarm signal generation circuit **225d**.

(73) When none of the control voltage detection circuit **225e**, the temperature detection circuit **225f** and the current detection circuit **225g** that subject the semiconductor element **21u** to detection of temperature and current, the temperature detection circuit and the current detection circuit (not shown) that subject the semiconductor element **21v** (see FIG. 1) to detection of temperature and current, and the temperature detection circuit and the current detection circuit (not shown) that subject the semiconductor element **21w** (see FIG. 1) to detection of temperature and current detects an abnormality, the alarm signal generation circuit **225d** outputs an output signal the signal level (i.e., a voltage level) of which is the low level. In contrast, when at least one of the control voltage detection circuit **225e**, the temperature detection circuit **225f** and the current detection circuit **225g** that subject the semiconductor element **21u** to detection of temperature and current, the temperature detection circuit and the current detection circuit that subject the semiconductor element **21v** to detection of temperature and current, and the temperature detection circuit and the current detection circuit that subject the semiconductor element **21w** to detection of temperature and current detects an abnormality, the alarm signal generation circuit **225d** outputs an output signal the signal level of which is the high level. Hereinafter, the control voltage detection circuit **225e**, the temperature detection circuit **225f** and the current detection circuit **225g** that subject the semiconductor element **21u** to detection of temperature and current, the temperature detection circuit and the current detection circuit that subject the semiconductor element **21v** to detection of temperature and current, and the temperature detection circuit and the current detection circuit that subject the semiconductor element **21w** to detection of temperature and current are sometimes referred to as “respective types of detection circuits”.

(74) The protection unit **225** includes a transistor **225b** having the gate G connected to the output terminal of the alarm signal generation circuit **225d** and a constant current source **225c** connected to the drain D of the transistor **225b**. The transistor **225b** is formed by, for example, an N-type MOSFET. The source S of the transistor **225b** is connected to the reference potential terminal T22.

(75) When none of the respective types of detection circuits detects an abnormality, the alarm signal generation circuit **225d** outputs an output signal the signal level of which is the low level to the gate G of the transistor **225b**. In contrast, when at least one of the respective types of detection circuits detects an abnormality, the alarm signal generation circuit **225d** outputs an output signal the signal level of which is the high level to the gate G of the transistor **225b** only for a predetermined period. Therefore, when the gate drive circuit **22A** changes from a normal state to an abnormal state, the alarm signal generation circuit **225d** outputs a pulse-shaped output signal the signal level of which is the high level to the gate G of the transistor **225b** only for a predetermined period.

(76) Thus, when the gate drive circuit **22A** is in the normal state, the transistor **225b** is put into the OFF state. In contrast, when the gate drive circuit **22A** changes from the normal state to an abnormal state, the transistor **225b** is put into the ON state only for a period during which the signal level of the output signal output from the alarm signal generation circuit **225d** is the high level. A case where the gate drive circuit **22A** is not in an abnormal state is a case where the temperature and current are a temperature and current at which the IGBT **211a** can operate and the control voltage input from the power source terminal **Tvin** has a voltage value at which the constant current supply units **221uA** to **223wA** can operate. Further, a case where the gate drive circuit **22A** is in an abnormal state is a case where the gate drive circuit **22A** is in at least one state of a state in which the IGBT **211a** is in a high-temperature condition or an overcurrent condition and a state in which the control voltage input from the power source terminal **Tvin** has a voltage value at which the constant current supply units **221uA** to **223wA** cannot operate.

(77) The protection unit **225** includes a resistance element **225h** connected between a connection point between the drain **D** of the transistor **225b** and the output terminal of the constant current source **225c** and the alarm signal output terminal **VFO**. One terminal of the resistance element **225h** is connected to the connection point between the drain **D** of the transistor **225b** and the output terminal of the constant current source **225c**. The other terminal of the resistance element **225h** is connected to the alarm signal output terminal **VFO**.

(78) Since the alarm signal output terminal **VFO** is connected to the drain **D** of the transistor **225b** via the resistance element **225h**, the alarm signal output terminal **VFO** is an open drain output terminal. When the gate drive circuit **22A** is in the normal state, since the signal level of the output signal output from the alarm signal generation circuit **225d** is the low level, the transistor **225b** is put into the OFF state. Thus, the voltage at the alarm signal output terminal **VFO** becomes a high level voltage. In contrast, when the gate drive circuit **22A** is in an abnormal state, since the signal level of the output signal output from the alarm signal generation circuit **225d** is the high level, the transistor **225b** is put into the ON state. Thus, the voltage at the alarm signal output terminal **VFO** becomes the potential of the reference potential terminal **T22** (for example, 0 V). As described above, the voltage at the alarm signal output terminal **VFO** becomes voltage obtained by inverting the voltage of the output signal output from the alarm signal generation circuit **225d**. As a result, the alarm signal **ALM** based on the output signal output from the alarm signal generation circuit **225d** is output from the alarm signal output terminal **VFO**.

(79) Although illustration is omitted, the protection unit **225uA** has the same configuration as the protection unit **225** and is configured to function in the same manner as the protection unit **225** except that the protection unit **225uA** detects an abnormality in only the control voltage of the constant current supply unit **221uA** and the semiconductor element **20u** that the constant current supply unit **221uA** drives. Further, the protection unit **225vA** has the same configuration as the protection unit **225** and is configured to function in the same manner as the protection unit **225** except that the protection unit **225vA** detects an abnormality in only the control voltage of the constant current supply unit **221vA** and the semiconductor element **20v** that the constant current supply unit **221vA** drives. Further, the protection unit **225wA** has the same configuration as the protection unit **225** and is configured to function in the same manner as the protection unit **225** except that the protection unit **225wA** detects an abnormality in only the control voltage of the constant current supply unit **221wA** and the semiconductor element **20w** that the constant current supply unit **221wA** drives.

(80) As illustrated in FIG. 2, the signal level determination unit **226uA** includes a plurality of (two in the present embodiment) comparators **226a** and **226b** configured to compare a comparison signal **Sc0** set to a predetermined signal level with the switching signal **Ssw4**. The signal level determination unit **226uA** includes a voltage generation unit **226c** configured to generate voltage of the comparison signal **Sc0**. The comparators **226a** and **226b** are respectively formed by, for example, operational amplifiers. The voltage generation unit **226c** is formed by, for example, a

constant voltage source.

(81) The non-inverting input terminal (+) of the comparator **226a** is connected to the switching signal input terminal Tsw4, the alarm signal output terminal VFO, and the other terminal of the resistance element **225h**. The inverting input terminal (−) of the comparator **226a** is connected to the positive electrode side of the voltage generation unit **226c**. The output terminal of the comparator **226a** is connected to the drive capability switching unit **224uA**. The non-inverting input terminal (+) of the comparator **226b** is connected to the positive electrode side of the voltage generation unit **226c**. The inverting input terminal (−) of the comparator **226b** is connected to the switching signal input terminal Tsw4, the alarm signal output terminal VFO, and the other terminal of the resistance element **225h**. The output terminal of the comparator **226b** is connected to the drive capability switching unit **224uA**. As described above, the comparator **226a** and the comparator **226b** have a relationship in which input destinations are reversed to each other. Thus, the comparator **226a** and the comparator **226b** (an example of at least two comparators among a plurality of comparators) output signals the signal levels of which are inverted to each other to the drive capability switching unit **224uA**.

(82) The negative electrode side of the voltage generation unit **226c** is connected to the reference potential terminal T22. The voltage generation unit **226c** is configured to generate voltage of the comparison signal Sco that serves as a reference for comparison in the comparators **226a** and **226b**. The Zener diode **6uL** that has a Zener voltage is connected to the switching signal input terminal Tsw4 in such a way that the signal level of the switching signal Ssw4 becomes higher or lower than the signal level of the comparison signal Sco. In the present embodiment, the voltage level of voltage that the voltage generation unit **226c** generates is set in advance. Thus, the Zener diode **6uL** that has a Zener voltage having a voltage level different from the voltage level of the voltage that the voltage generation unit **226c** generates is connected to the switching signal input terminal Tsw4 in such a way that the drive capability of all the constant current supply units including the constant current supply unit **223uA** that are included in the gate drive circuit **22A** coincides with a desired drive capability of a user of the intelligent power module **2A**. The intelligent power module **2A** is configured such that the user of the intelligent power module **2A** can connect the Zener diode **6uL** to the switching signal input terminal Tsw4.

(83) When a Zener diode **6uL** having a Zener voltage higher than the voltage generated by the voltage generation unit **226c** is connected to the switching signal input terminal Tsw4, the signal level of the switching signal Ssw4 becomes higher than the signal level of the comparison signal Sco. Therefore, in this case, the comparator **226a** outputs an output signal Soa the signal level of which is the high level to the drive capability switching unit **224uA**, and the comparator **226b** outputs an output signal Sob the signal level of which is the low level to the drive capability switching unit **224uA**. In contrast, when a Zener diode **6uL** having a Zener voltage lower than the voltage generated by the voltage generation unit **226c** is connected to the switching signal input terminal Tsw4, the signal level of the switching signal Ssw4 becomes lower than the signal level of the comparison signal Sco. Therefore, in this case, the comparator **226a** outputs the output signal Soa the signal level of which is the low level to the drive capability switching unit **224uA**, and the comparator **226b** outputs the output signal Sob the signal level of which is the high level to the drive capability switching unit **224uA**.

(84) As illustrated in FIG. 2, the drive capability switching unit **224uA** includes a negative AND (NAND) gate **224a** (an example of a first operation unit) that performs a logical operation on the signal level of the output signal Soa of the comparator **226a** (an example of one of two comparators) and the signal level of the output signal So0 (an example of a signal based on an input signal) of the comparator **223a**. In addition, the drive capability switching unit **224uA** includes a negative AND (NAND) gate **224b** (an example of a second operation unit) that performs a logical operation on the signal level of the output signal Sob of the comparator **226b** (an example of the other of the two comparators) and the signal level of the output signal So0.

(85) One input terminal of the NAND gate **224a** is connected to the output terminal of the comparator **223a** arranged in the constant current supply unit **223uA**. The other input terminal of the NAND gate **224a** is connected to the output terminal of the comparator **226a** arranged in the signal level determination unit **226uA**. The output terminal of the NAND gate **224a** is connected to the gate G of the transistor **223f** arranged in the constant current supply unit **223uA**. Thus, the NAND gate **224a** outputs an operation result signal **Scaa** to the gate G (an example of a control signal input terminal of one of two transistors) of the transistor **223f**.

(86) One input terminal of the NAND gate **224b** is connected to the output terminal of the comparator **223a** arranged in the constant current supply unit **223uA**. The other input terminal of the NAND gate **224b** is connected to the output terminal of the comparator **226b** arranged in the signal level determination unit **226uA**. The output terminal of the NAND gate **224b** is connected to the gate G of the transistor **223g** arranged in the constant current supply unit **223uA**. Thus, the NAND gate **224b** outputs an operation result signal **Scab** to the gate G (an example of a control signal input terminal of the other of the two transistors) of the transistor **223g**.

(87) To the other input terminals of the respective ones of the NAND gates **224a** and **224b**, the output signal **So0** is input. On the other hand, to one input terminals of the respective ones of the NAND gates **224a** and **224b**, the output signals **Soa** and **Sob** the signal levels of which are inverted to each other are input, respectively. Thus, when the signal level of the output signal **So0** is the high level, the NAND gates **224a** and **224b** output the operation result signals **Scaa** and **Scab** the signal levels of which are inverted to each other to the gates G of the respective ones of the transistors **223f** and **223g**. By way of this, when the transistor **223f** switches to the ON state, the transistor **223g** maintains the OFF state, and, when the transistor **223g** switches to the ON state, the transistor **223f** maintains the OFF state.

(88) The output signal **So0** output from the comparator **223a** is a signal based on the input signal **SinUL**. Thus, the NAND gates **224a** and **224b** that perform a logical operation using the signal level of the output signal **So0** are configured to perform a logical operation, based on the signal level of the input signal **SinUL**. Therefore, the drive capability switching unit **224uA** controls at least one of the transistors **223f** and **223g** to the ON state, based on the determination results (i.e., the signal levels of the output signals **Soa** and **Sob**) in the signal level determination unit **226uA** and the signal level of the input signal **SinUL**. In the present embodiment, the drive capability switching unit **224uA** controls either of the transistors **223f** and **223g** to the ON state.

(89) Note that, in the drive capability switching units **222uA**, **222vA**, and **222wA** for driving the semiconductor elements **20u**, and **20w** that constitute the upper arm portion, voltage level shift circuits may be connected to the input terminals of the NAND gates **224a** and **224b**. The output signals **So0**, **Soa**, and **Sob** are input to the voltage level shift circuits. By way of this, the voltage levels of the output signals **So0**, **Soa**, and **Sob** are converted to voltage levels matching the upper arm portion and subsequently are logically operated by the NAND gates **224a** and **224b**.

(90) Further, in the constant current supply units **221uA**, **221vA**, and **221wA** for driving the semiconductor elements **20u**, and **20w** that constitute the upper arm portion, the sources S of the respective ones of the transistors **223f** and **223g** are individually connected to the power source terminals of the respective ones of the U-phase, the V-phase, and the W-phase of the upper arm portion.

(91) (Operation of Drive Circuit of Switching Element)

(92) Next, operation of the gate drive circuit **22A** serving as the drive circuit of switching elements according to the present embodiment will be described using FIGS. **1** and **2**. The constant current supply units **221uA** to **223wA** operate in the same manner as one another. Thus, operation of the constant current supply units **221uA** to **223wA** will be described using the constant current supply unit **223uA** as an example. The drive capability switching units **222uA** to **224wA** operate in the same manner as one another. Thus, operation of the drive capability switching units **222uA** to **224wA** will be described using the drive capability switching unit **224uA** as an example.

(93) Table 1 is a truth table of the transistors **223f** and **223g** arranged in the constant current supply unit **223uA**. “SinUL” in Table 1 indicates the input signal SinUL input to the constant current supply unit **223uA** from the control device **3**. “So0” in Table 1 indicates the output signal So0 that the comparator **223a** arranged in the constant current supply unit **223uA** outputs. “Soa” in Table 1 indicates the output signal Soa that the comparator **226a** arranged in the signal level determination unit **226uA** outputs. “Sob” in Table 1 indicates the output signal Sob that the comparator **226b** arranged in the signal level determination unit **226uA** outputs.

(94) “Scaa” in Table 1 indicates the operation result signal Scaa that the NAND gate **224a** arranged in the drive capability switching unit **224uA** outputs. “Scab” in Table 1 indicates the operation result signal Scab that the NAND gate **224b** arranged in the drive capability switching unit **224uA** outputs. “Qf” in Table 1 indicates the transistor **223f** arranged in the constant current supply unit **223uA**. “Qg” in Table 1 indicates the transistor **223g** arranged in the constant current supply unit **223uA**. “L” in Table 1 indicates that the signal level is the low level. “H” in Table 1 indicates that the signal level is the high level. “ON” in Table 1 indicates that the transistor is in the ON state. “OFF” in Table 1 indicates that the transistor is in the OFF state.

(95) TABLE-US-00001

SinUL	So0	Soa	Sob	Scaa	Scab	Qf	Qg	Ssw4 > Sco	Sco	H	L	H	L	H	H
OFF	OFF	L	H	H	L	L	H	ON	OFF	Ssw4 < Sco	H	L	L	H	H
OFF	OFF	L	H	L	L	H	ON	OFF	OFF	OFF	L	H	L	H	L
OFF	ON	L	H	L	L	H	ON	OFF	OFF	OFF	L	H	L	H	L

 (Case where Ssw4>Sco)

(96) First, operation of the gate drive circuit **22A** in the case where the Zener voltage of the Zener diode **6uL** is higher than the voltage generated by the voltage generation unit **226c** arranged in the signal level determination unit **226uA**, that is, in the case where the signal level of the switching signal Ssw4 is higher than the signal level of the comparison signal Sco, will be described.

(97) When the input signal SinUL the signal level of which is the high level is input from the control device **3** to the constant current supply unit **223uA**, the signal level of the output signal So0 output from the comparator **223a** becomes the low level, as shown in the upper row in the row “Ssw4>Sco” in table 1.

(98) When the output signal So0 the signal level of which is the low level is input to the other input terminal of the NAND gate **224a** arranged in the drive capability switching unit **224uA**, the NAND gate **224a** outputs the operation result signal Scaa the signal level of which is the high level, regardless of the signal level of the output signal Soa input to one input terminal, as shown in Table 1. Likewise, when the output signal So0 the signal level of which is the low level is input to the other input terminal of the NAND gate **224b** arranged in the drive capability switching unit **224uA**, the NAND gate **224b** outputs the operation result signal Scab the signal level of which is the high level, regardless of the signal level of the output signal Sob input to one input terminal, as shown in Table 1.

(99) As a result, as shown in the upper row in the row “Ssw4>Sco” in Table 1, both transistors **223f** and **223g** arranged in the constant current supply unit **223uA** are controlled to the OFF state. Thus, when the IGBT **211a** arranged in the semiconductor element **21u** is in the OFF state, the constant current supply unit **223uA** drives the IGBT **211a** to maintain the OFF state. In contrast, when the IGBT **211a** is in the ON state, the constant current supply unit **223uA** drives the IGBT **211a** to switch from the ON state to the OFF state. That is, the constant current supply unit **223uA** drives the IGBT **211a** to turn off.

(100) When the input signal SinUL the signal level of which is the low level is input from the control device **3** to the constant current supply unit **223uA**, the signal level of the output signal So0 output from the comparator **223a** becomes the high level, as shown in the lower row in the row “Ssw4>Sco” in Table 1.

(101) Since the signal level of the switching signal Ssw4 is higher than the signal level of the comparison signal Sco, the comparator **226a** arranged in the signal level determination unit **226uA** outputs the output signal Soa the signal level of which is the high level, as shown in the lower row in the row “Ssw4>Sco” in Table 1. When the output signal Soa the signal level of which is the high

level is input to one input terminal of the NAND gate **224a** and the output signal **So0** the signal level of which is the high level is input to the other input terminal of the NAND gate **224a**, the NAND gate **224a** outputs the operation result signal **Scaa** the signal level of which is the low level to the gate G of the transistor **223f**, as shown in the lower row in the row “**Ssw4>Sco**” in Table 1.

(102) On the other hand, as shown in the lower row in the row “**Ssw4>Sco**” in Table 1, the comparator **226b** arranged in the signal level determination unit **226uA** outputs the output signal **Sob** the signal level of which is the low level. When the output signal **Sob** the signal level of which is the low level is input to one input terminal of the NAND gate **224b** and the output signal **So0** the signal level of which is the high level is input to the other input terminal of the NAND gate **224b**, the NAND gate **224b** outputs the operation result signal **Scab** the signal level of which is the high level to the gate G of the transistor **223g**, as shown in the lower row in the row “**Ssw4>Sco**” in Table 1.

(103) As a result, as shown in the lower row in the row “**Ssw4>Sco**” in Table 1, the transistor **223f** is controlled to the ON state and the transistor **223g** is controlled to the OFF state. Thus, when the IGBT **211a** arranged in the semiconductor element **21u** is in the OFF state, the constant current supply unit **223uA** supplies the IGBT **211a** with constant current from the transistor **223f** that has a larger current amount of constant current between the transistors **223f** and **223g**. By way of this, the constant current supply unit **223uA** causes the IGBT **211a** to switch to the ON state with high drive capability.

(104) In contrast, when the IGBT **211a** is in the ON state, the constant current supply unit **223uA** drives the IGBT **211a** to maintain the ON state by supplying constant current from the transistor **223f** to the IGBT **211a**.

(105) (Case where **Ssw4<Sco**)

(106) Next, operation of the gate drive circuit **22A** in the case where the Zener voltage of the Zener diode **6uL** is lower than the voltage generated by the voltage generation unit **226c** arranged in the signal level determination unit **226uA**, that is, in the case where the signal level of the switching signal **Ssw4** is lower than the signal level of the comparison signal **Sco**, will be described.

(107) The drive capability switching unit **224uA** operates in the same manner between the case where the signal level of the switching signal **Ssw4** is lower than the signal level of the comparison signal **Sco** and the case where the signal level of the switching signal **Ssw4** is higher than the signal level of the comparison signal **Sco**. Thus, as shown in the upper row in the row “**Ssw4<Sco**” in Table 1, both transistors **223f** and **223g** are controlled to the OFF state. Therefore, the constant current supply unit **223uA** drives the IGBT **211a** arranged in the semiconductor element **21u** in the same manner as in the case where the signal level of the switching signal **Ssw4** is higher than the signal level of the comparison signal **Sco**.

(108) When the input signal **SinUL** the signal level of which is the low level is input from the control device **3** to the constant current supply unit **223uA**, the signal level of the output signal **So0** output from the comparator **223a** becomes the high level, as shown in the lower row in the row “**Ssw4<Sco**” in Table 1.

(109) Since the signal level of the switching signal **Ssw4** is lower than the signal level of the comparison signal **Sco**, the comparator **226a** outputs the output signal **Soa** the signal level of which is the low level, as shown in the lower row in the row “**Ssw4<Sco**” in Table 1. When the output signal **Soa** the signal level of which is the low level is input to one input terminal of the NAND gate **224a** and the output signal **So0** the signal level of which is the high level is input to the other input terminal of the NAND gate **224a**, the NAND gate **224a** outputs the operation result signal **Scaa** the signal level of which is the high level to the gate G of the transistor **223f**, as shown in the lower row in the row “**Ssw4<Sco**” in Table 1.

(110) On the other hand, as shown in the lower row in the row “**Ssw4<Sco**” in Table 1, the comparator **226b** outputs the output signal **Sob** the signal level of which is the high level. When the output signal **Sob** the signal level of which is the high level is input to one input terminal of the

NAND gate **224b** and the output signal **So0** the signal level of which is the high level is input to the other input terminal of the NAND gate **224b**, the NAND gate **224b** outputs the operation result signal **Scab** the signal level of which is the low level to the gate **G** of the transistor **223g**, as shown in the lower row in the row “**Ssw4<Sco**” in Table 1.

(111) As a result, as shown in the lower row in the row “**Ssw4<Sco**” in Table 1, the transistor **223f** is controlled to the OFF state and the transistor **223g** is controlled to the ON state. Thus, when the IGBT **211a** arranged in the semiconductor element **21u** is in the OFF state, the constant current supply unit **223uA** supplies the IGBT **211a** with constant current from the transistor **223g** that has a smaller current amount of constant current between the transistors **223f** and **223g**. By way of this, the constant current supply unit **223uA** causes the IGBT **211a** to switch to the ON state with low drive capability.

(112) In contrast, when the IGBT **211a** is in the ON state, the constant current supply unit **223uA** drives the IGBT **211a** to maintain the ON state by supplying constant current from the transistor **223g** to the IGBT **211a**.

(113) As described above, the gate drive circuit **22A** and the intelligent power module **2A** are capable of changing drive capability to drive the IGBTs **201a** arranged in the semiconductor elements **20u**, **20v**, and **20w** and the IGBTs **211a** arranged in the semiconductor elements **21u**, **21v**, and **21w** only by connecting the Zener diodes **6uH**, **6vH**, **6wH**, **6uL**, **6vL**, and **6wL** having predetermined Zener voltages to the switching signal input terminals **Tsw1**, **Tsw2**, **Tsw3**, **Tsw4**, **Tsw5**, and **Tsw6**, respectively.

(114) As described in the foregoing, the gate drive circuit **22A** as a drive circuit of switching elements according to the present embodiment, in order to cause the IGBTs **201a** and **211a** of the semiconductor elements **20u** to **21w** to switch to the ON state according to the signal levels of the input signals **SinUH** to **Sin WL** input from the control device **3**, includes the constant current supply units **221uA** to **223wA** configured to supply constant currents to the gates **G** of the IGBTs **201a** and **211a**, the switching signal input terminals **Tsw1** to **Tsw6** to which the switching signals **Ssw1** to **Ssw6** for switching the drive capabilities of the constant current supply units **221uA** to **223wA** causing the IGBTs **201a** and **211a** to operate are input, the signal level determination units **227uA**, **227vA**, **227wA**, **226uA**, **226vA**, and **226wA** configured to determine the signal levels of the switching signals **Ssw1** to **Ssw6**, and the drive capability switching units **222uA** to **224wA** configured to change current amounts of constant currents output from the constant current supply units **221uA** to **223wA**, based on determination results in the signal level determination units **227uA**, **227vA**, **227wA**, **226uA**, **226vA**, and **226wA** and the signal levels of the input signals **SinUH** to **Sin WL** and thereby switch the drive capabilities of constant current supply units **221uA** to **223wA**, respectively.

(115) By way of this, the gate drive circuit **22A** is capable of selecting a drive capability to drive the switching elements in a simple method.

(116) The intelligent power module **2A** according to the present embodiment includes the IGBTs **201a** and **211a** and the gate drive circuit **22A**.

(117) By way of this, the intelligent power module **2A** is capable of selecting a drive capability to drive the switching elements in a simple method.

(118) (Variation)

(119) Although, in the present embodiment, in each of the constant current supply units **221uA** to **223wA** arranged in the gate drive circuit **22A**, current amounts of constant currents supplied to an IGBT **201a** or **211a** from the transistors **223f** and **223g** are different from each other, the current amounts of the constant currents may be the same. In this case, when the signal level of the output signal **Soa** input from the comparator **226a** arranged in each of the signal level determination units **227uA**, **227vA**, **227wA**, **226uA**, **226vA**, **226wA** is the high level and the signal level of the output signal **Sob** output from the comparator **226b** arranged in the signal level determination unit is the low level, a corresponding one of the drive capability switching units **222uA** to **224wA** controls

both transistors **223f** and **223g** to the ON state. On the other hand, when the signal level of the output signal Soa input from the comparator **226a** is the low level and the signal level of the output signal Sob output from the comparator **226b** is the high level, the corresponding one of the drive capability switching units **222uA** to **224wA** controls one of the transistors **223f** and **223g** to the ON state and controls the other to the OFF state.

(120) When both transistors **223f** and **223g** are in the ON state, the constant current supply unit is capable of supplying the IGBT **201a** or **211a** with a double amount of constant current compared with a case where one of the transistors **223f** and **223g** is in the ON state. By way of this, the gate drive circuit **22A** and the intelligent power module **2A** are capable of changing drive capability to drive the IGBTs **201a** and **211a** only by connecting the Zener diodes **6uH** to **6wL** having predetermined Zener voltages to the switching signal input terminals Tsw1 to Tsw6, respectively.

Second Embodiment

(121) A drive circuit of a switching element and an intelligent power module according to a second embodiment of the present invention will be described using FIG. 3. The drive circuit of a switching element and the intelligent power module according to the present embodiment has a feature that a constant current supply unit includes three transistors. In the following description of the drive circuit of switching elements and the intelligent power module according to the present embodiment, constituent elements that achieve the same effects and the same functions as those of the constituent elements in the gate drive circuit **22A** and the intelligent power module **2A** serving as the drive circuit of switching elements according to the above-described first embodiment will be denoted by the same reference numerals and description thereof will be omitted

(122) (Configuration of Intelligent Power Module)

(123) Since an overall configuration of an intelligent power module **2B** according to the present embodiment is the same as that of the intelligent power module **2A** according to the above-described first embodiment, an illustration and description thereof will be omitted. In the following description of the intelligent power module **2B**, FIG. 1 will be referred to as needed.

(124) (Configuration of Drive Circuit of Switching Element)

(125) Next, a schematic configuration of a gate drive circuit **22B** serving as the drive circuit of switching elements according to the present embodiment will be described using FIG. 3. Constant current supply units corresponding to respective ones of a U-phase, a V-phase, and a W-phase of an upper arm and a U-phase, a V-phase, and a W-phase of a lower arm arranged in the gate drive circuit **22B** have the same configuration as one another. Thus, the configurations of the constant current supply units will be described using a constant current supply unit **223uB** (corresponding to the constant current supply unit **223uA** in the above-described first embodiment) corresponding to the U-phase of the lower arm as an example. Drive capability switching units corresponding to respective ones of the U-phase, the V-phase, and the W-phase of the upper arm and the U-phase, the V-phase, and the W-phase of the lower arm arranged in the gate drive circuit **22B** have the same configuration as one another. Thus, the configurations of the drive capability switching units will be described using a drive capability switching unit **224uB** (corresponding to the drive capability switching unit **224uA** in the above-described first embodiment) corresponding to the U-phase of the lower arm as an example. Signal level determination units corresponding to respective ones of the U-phase, the V-phase, and the W-phase of the upper arm and the U-phase, the V-phase, and the W-phase of the lower arm arranged in the gate drive circuit **22B** have the same configuration as one another. Thus, the configurations of the signal level determination units will be described using a signal level determination unit **226uB** (corresponding to the signal level determination unit **226uA** in the above-described first embodiment) corresponding to the U-phase of the lower arm as an example.

(126) FIG. 3 is a circuit block diagram illustrative of a schematic configuration of the gate drive circuit **22B**. In FIG. 3, among the constituent components arranged in the gate drive circuit **22B**, the constant current supply unit **223uB**, the signal level determination unit **226uB**, and the drive

capability switching unit **224uB** are illustrated. In addition, in FIG. 3, to facilitate understanding, a semiconductor element **21u** including an IGBT **211a** to which the constant current supply unit **223uB** supplies current, a Zener diode **6uL** connected to a switching signal input terminal **Tsw4**, and a power source **8L** connected to a power source terminal **Tvin** are also illustrated. Note that, since protection units that are arranged in the gate drive circuit **22B** and correspond to the U-phases, the V-phases, and the W-phases of the respective ones of the upper arm and the lower arm have the same configurations and function in the same manner as the protection units **225uA**, **225vA**, **225wA**, and **225** arranged in the gate drive circuit **22A** according to the above-described first embodiment, an illustration and a description thereof in FIG. 3 will be omitted.

(127) As illustrated in FIG. 3, the constant current supply unit **223uB** includes a plurality of (three in the present embodiment) transistors **223f**, **223g**, and **223h** configured to supply constant current to the gate G of the IGBT **211a** arranged in the semiconductor element **21u**. The transistors **223f**, **223g**, and **223h** are respectively formed by, for example, P-type MOSFETs. The source S of the transistor **223f** and the sources S of the respective ones of the transistor **223g** and the transistor **223h** are connected to the power source terminal **Tvin**. The drains D of the respective ones of the transistors **223f**, **223g**, and **223h** are connected to the gate G of the IGBT **211a** arranged in the semiconductor element **21u**. The gates G of the respective ones of the transistors **223f**, **223g**, and **223h** are connected to the drive capability switching unit **224uB**.

(128) Current amounts of constant currents that the transistors **223f**, **223g**, and **223h** (an example of at least two transistors among a plurality of transistors) respectively supplies to the gate G of the IGBT **211a** arranged in the semiconductor element **21u** are different from one another. For example, the transistor **223f** has a larger transistor size than the transistors **223g** and **223h**. The transistor **223g** has a larger transistor size than the transistor **223h**. By way of this, the current amount of the constant current from the transistor **223f** is larger than the current amounts of the constant currents from the respective ones of the transistors **223g** and **223h**. In addition, the current amount of the constant current from the transistor **223g** is larger than the current amount of the constant current from the transistor **223h**.

(129) Although details will be described later, in the case where the transistor **223f** is selected as a transistor to supply constant current to the IGBT **211a** arranged in the semiconductor element **21u**, the constant current supply unit **223uB** is capable of causing the IGBT **211a** to switch to the ON state with higher drive capability than a case where the transistor **223g** or **223h** is selected. In addition, in the case where the transistor **223g** is selected as a transistor to supply constant current to the IGBT **211a** arranged in the semiconductor element **21u**, the constant current supply unit **223uB** is capable of causing the IGBT **211a** to switch to the ON state with higher drive capability than a case where the transistor **223h** is selected. Further, in the case where the transistor **223h** is selected as a transistor to supply constant current to the IGBT **211a** arranged in the semiconductor element **21u**, the constant current supply unit **223uB** is capable of causing the IGBT **211a** to switch to the ON state with the lowest drive capability. As described above, the constant current supply unit **223uB** is capable of, by changing a transistor to be caused to operate among the transistors **223f**, **223g**, and **223h**, switching drive capability to drive the IGBT **211a**.

(130) As illustrated in FIG. 3, the signal level determination unit **226uB** includes a plurality of (four in the present embodiment) comparators **226a**, **226b**, **226d**, and **226e** configured to compare a comparison signal set to a predetermined signal level with a switching signal **Ssw4**. The comparators **226a** and **226b** compare a comparison signal **Scoc** and the switching signal **Ssw4** with each other, and the comparators **226d** and **226e** compare a comparison signal **Scof** and the switching signal **Ssw4** with each other. The signal level determination unit **226uB** includes a voltage generation unit **226c** configured to generate voltage of the comparison signal **Scoc** and a voltage generation unit **226f** configured to generate voltage of the comparison signal **Scof**. The comparators **226a**, **226b**, **226d**, and **226e** are respectively formed by, for example, operational amplifiers. The voltage generation units **226c** and **226f** are respectively formed by, for example,

constant voltage sources.

(131) The non-inverting input terminal (+) of the comparator **226a** is connected to the switching signal input terminal Tsw4. The inverting input terminal (−) of the comparator **226a** is connected to the positive electrode side of the voltage generation unit **226c**. The output terminal of the comparator **226a** is connected to the drive capability switching unit **224uB**. The non-inverting input terminal (+) of the comparator **226b** is connected to the positive electrode side of the voltage generation unit **226c**. The inverting input terminal (−) of the comparator **226b** is connected to the switching signal input terminal Tsw4. The output terminal of the comparator **226b** is connected to the drive capability switching unit **224uB**.

(132) The non-inverting input terminal (+) of the comparator **226d** is connected to the switching signal input terminal Tsw4. The inverting input terminal (−) of the comparator **226d** is connected to the positive electrode side of the voltage generation unit **226f**. The output terminal of the comparator **226d** is connected to the drive capability switching unit **224uB**. The non-inverting input terminal (+) of the comparator **226e** is connected to the positive electrode side of the voltage generation unit **226f**. The inverting input terminal (−) of the comparator **226e** is connected to the switching signal input terminal Tsw4. The output terminal of the comparator **226e** is connected to the drive capability switching unit **224uB**.

(133) As described above, the comparator **226a** and the comparator **226b** have a relationship in which input destinations are reversed to each other. Thus, the comparator **226a** and the comparator **226b** (an example of at least two comparators among a plurality of comparators) output signals the signal levels of which are inverted to each other to the drive capability switching unit **224uB**. In addition, the comparator **226d** and the comparator **226e** have a relationship in which input destinations are reversed to each other. Thus, the comparator **226d** and the comparator **226e** (an example of at least two comparators among a plurality of comparators) output output signals the signal levels of which are inverted to each other to the drive capability switching unit **224uB**.

(134) The negative electrode side of the voltage generation unit **226c** is connected to the reference potential terminal T22. The voltage generation unit **226c** is configured to generate voltage of the comparison signal Scoc that serves as a reference for comparison in the comparators **226a** and **226b**. The negative electrode side of the voltage generation unit **226f** is connected to the reference potential terminal T22. The voltage generation unit **226f** is configured to generate voltage of the comparison signal Scof that serves as a reference for comparison in the comparators **226d** and **226e**. The voltage that the voltage generation unit **226c** generates is set to be, for example, higher than the voltage that the voltage generation unit **226f** generates. Therefore, the signal level of the comparison signal Scoc is higher than the signal level of the comparison signal Scof.

(135) The Zener diode **6uL** that has a Zener voltage causing the signal level of the switching signal Ssw4 to be higher than the signal level of the comparison signal Scoc, a Zener voltage causing the signal level of the switching signal Ssw4 to be lower than the comparison signal Scoc and higher than the comparison signal Scof, or a Zener voltage causing the signal level of the switching signal Ssw4 to be lower than the signal level of the comparison signal Scof is connected to the switching signal input terminal Tsw4. In the present embodiment, the voltage levels of voltages that the voltage generation units **226c** and **226f** generate are set in advance. Thus, the Zener diode **6uL** that has a Zener voltage having a voltage level different from the voltage levels of voltages that the voltage generation units **226c** and **226f** generate is connected to the switching signal input terminal Tsw4 in such a way that the drive capability of all the constant current supply units including the constant current supply unit **223uB** that are included in the gate drive circuit **22B** coincides with a desired drive capability of a user of the intelligent power module **2B**. The intelligent power module **2B** is configured such that the user of the intelligent power module **2B** can connect the Zener diode **6uL** to the switching signal input terminal Tsw4.

(136) When a Zener diode **6uL** having a Zener voltage higher than the voltages generated by the voltage generation units **226c** and **226f** is connected to the switching signal input terminal Tsw4,

the signal level of the switching signal Ssw4 becomes higher than the signal levels of the respective ones of the comparison signals Scoc and Scof. Therefore, in this case, the comparator 226a outputs an output signal Soa the signal level of which is the high level to the drive capability switching unit 224uB, and the comparator 226b outputs an output signal Sob the signal level of which is the low level to the drive capability switching unit 224uB. In addition, in this case, the comparator 226d outputs an output signal Sod the signal level of which is the high level to the drive capability switching unit 224uB, and the comparator 226e outputs an output signal Soe the signal level of which is the low level to the drive capability switching unit 224uB.

(137) Further, when a Zener diode 6uL having a Zener voltage lower than the voltage generated by the voltage generation unit 226c and higher than the voltage generated by the voltage generation unit 226f is connected to the switching signal input terminal Tsw4, the signal level of the switching signal Ssw4 becomes lower than the signal level of the comparison signal Scoc and higher than the signal level of the comparison signal Scof. Therefore, in this case, the comparator 226a outputs the output signal Soa the signal level of which is the low level to the drive capability switching unit 224uB, and the comparator 226b outputs the output signal Sob the signal level of which is the high level to the drive capability switching unit 224uB. Further, in this case, the comparator 226d outputs the output signal Sod the signal level of which is the high level to the drive capability switching unit 224uB, and the comparator 226e outputs the output signal Soe the signal level of which is the low level to the drive capability switching unit 224uB.

(138) When a Zener diode 6uL having a Zener voltage lower than the voltages generated by the voltage generation units 226c and 226f is connected to the switching signal input terminal Tsw4, the signal level of the switching signal Ssw4 becomes lower than the signal levels of the respective ones of the comparison signals Scoc and Scof. Therefore, in this case, the comparator 226a outputs the output signal Soa the signal level of which is the low level to the drive capability switching unit 224uB, and the comparator 226b outputs the output signal Sob the signal level of which is the high level to the drive capability switching unit 224uB. In addition, in this case, the comparator 226d outputs the output signal Sod the signal level of which is the low level to the drive capability switching unit 224uB, and the comparator 226e outputs the output signal Soe the signal level of which is the high level to the drive capability switching unit 224uB.

(139) As illustrated in FIG. 3, the drive capability switching unit 224uB includes a logical AND (AND) gate 224d that performs a logical operation on the signal level of the output signal Sob of the comparator 226b and the signal level of the output signal Sod of the comparator 226d. The drive capability switching unit 224uB includes a negative AND (NAND) gate 224a (an example of a first operation unit) that performs a logical operation on the signal level of the output signal Soa of the comparator 226a (an example of one of two comparators) and the signal level of an output signal So0 (an example of a signal based on an input signal) of a comparator 223a. In addition, the drive capability switching unit 224uB includes a negative AND (NAND) gate 224b (an example of a second operation unit) that performs a logical operation on the signal level of an operation result signal Scad that the AND gate 224d outputs and the signal level of the output signal So0. In this configuration, the comparator 226b is equivalent to an example of the other of the two comparators. Thus, the operation result signal Scad that is obtained by performing a logical operation using the output signal Sob of the comparator 226b is equivalent to an example of a signal based on the output signal Sob of the comparator 226b. Further, the drive capability switching unit 224uB includes a negative AND (NAND) gate 224c that performs a logical operation on the signal level of the output signal Soe of the comparator 226e and the signal level of the output signal So0 of the comparator 223a.

(140) One input terminal of the NAND gate 224a is connected to the output terminal of the comparator 223a arranged in the constant current supply unit 223uB. The other input terminal of the NAND gate 224a is connected to the output terminal of the comparator 226a arranged in the signal level determination unit 226uB. The output terminal of the NAND gate 224a is connected to

the gate G of the transistor **223f** arranged in the constant current supply unit **223uB**. Thus, the NAND gate **224a** outputs an operation result signal Scaa to the gate G (an example of a control signal input terminal of one of two transistors) of the transistor **223f**.

(141) One input terminal of the NAND gate **224b** is connected to the output terminal of the comparator **223a** arranged in the constant current supply unit **223uB**. The other input terminal of the NAND gate **224b** is connected to the output terminal of the AND gate **224d**. The output terminal of the NAND gate **224b** is connected to the gate G of the transistor **223g** arranged in the constant current supply unit **223uB**. Thus, the NAND gate **224b** outputs an operation result signal Scab to the gate G (an example of a control signal input terminal of the other of the two transistors) of the transistor **223g**.

(142) One input terminal of the NAND gate **224c** is connected to the output terminal of the comparator **223a** arranged in the constant current supply unit **223uB**. The other input terminal of the NAND gate **224c** is connected to the output terminal of the comparator **226e** arranged in the signal level determination unit **226uB**. The output terminal of the NAND gate **224c** is connected to the gate G of the transistor **223h** arranged in the constant current supply unit **223uB**. Thus, the NAND gate **224c** outputs an operation result signal Scac to the gate G of the transistor **223h**.

(143) To the other input terminals of the respective ones of the NAND gates **224a**, **224b**, and **224c**, the output signal So0 is input. Thus, the signal levels of the respective ones of the operation result signals Scaa, Scab, and Scac depend on the signal levels of signals input to one input terminals of the respective ones of the NAND gates **224a**, **224b**, and **224c**, respectively. Although details will be described later, depending on a high/low relationship among the signal level of the switching signal Ssw4, the signal level of the comparison signal Scoc, and the signal level of the comparison signal Scof, only one of the operation result signals Scaa, Scab, and Scac becomes the low level. By way of this, when the transistor **223f** switches to the ON state, the transistor **223g** and the transistor **223h** maintain the OFF state. In addition, when the transistor **223g** switches to the ON state, the transistor **223f** and the transistor **223h** maintain the OFF state. Further, when the transistor **223h** switches to the ON state, the transistor **223f** and the transistor **223g** maintain the OFF state.

(144) The output signal So0 output from the comparator **223a** is a signal based on an input signal SinUL. Thus, the NAND gates **224a**, **224b**, and **224c** that perform a logical operation using the signal level of the output signal So0 are configured to perform a logical operation, based on the signal level of the input signal SinUL. Therefore, the drive capability switching unit **224uB** controls at least one of the transistors **223f** and **223g** to the ON state, based on the determination results (i.e., the signal levels of the output signals Soa, Sob, Sod, and Soe) in the signal level determination unit **226uB** and the signal level of the input signal SinUL. In the present embodiment, the drive capability switching unit **224uB** controls one of the transistors **223f**, **223g**, and **223h** to the ON state.

(145) Note that, in the drive capability switching units for driving the semiconductor elements **20u**, **20v**, and **20w** that constitute the upper arm portion (corresponding to the drive capability switching units **222uA**, **222vA**, and **222wA** in the above-described first embodiment), voltage level shift circuits may be connected to the input terminals of the NAND gates **224a**, **224b**, and **224c** and the AND gate **224d**. The output signals So0, Soa, Sob, Sod, and Soe are input to the voltage level shift circuits. By way of this, the voltage levels of the output signals So0, Soa, Sob, Sod, and Soe are converted to voltage levels matching the upper arm portion and subsequently are logically operated by the NAND gates **224a**, **224b**, and **224c** and the AND gate **224d**.

(146) Further, in the constant current supply units for driving the semiconductor elements **20u**, **20v**, and **20w** that constitute the upper arm portion (corresponding to the constant current supply units **221uA**, **221vA**, and **221wA** in the above-described first embodiment), the sources S of the respective ones of the transistors **223f**, **223g**, and **223h** are individually connected to the power source terminals of the respective ones of the U-phase, the V-phase, and the W-phase of the upper arm portion.

(147) (Operation of Drive Circuit of Switching Element)

(148) Next, operation of the gate drive circuit **22B** serving as the drive circuit of switching elements according to the present embodiment will be described using FIG. 3. The constant current supply units corresponding to the respective ones of the U-phase, the V-phase, and the W-phase of the upper arm and the U-phase, the V-phase, and the W-phase of the lower arm arranged in the gate drive circuit **22B** operate in the same manner as one another. Thus, operation of the constant current supply units will be described using the constant current supply unit **223uB** corresponding to the U-phase of the lower arm as an example. The drive capability switching units corresponding to the respective ones of the U-phase, the V-phase, and the W-phase of the upper arm and the U-phase, the V-phase, and the W-phase of the lower arm arranged in the gate drive circuit **22B** operate in the same manner as one another. Thus, operation of the drive capability switching units will be described using the drive capability switching unit **224uB** corresponding to the U-phase of the lower arm as an example. The signal level determination units corresponding to the respective ones of the U-phase, the V-phase, and the W-phase of the upper arm and the U-phase, the V-phase, and the W-phase of the lower arm arranged in the gate drive circuit **22B** operate in the same manner as one another. Thus, operation of the signal level determination units will be described using the signal level determination unit **226uB** corresponding to the U-phase of the lower arm as an example.

(149) Table 2 is a truth table of the transistors **223f**, **223g**, and **223h** arranged in the constant current supply unit **223uB**. “SinUL” in Table 2 indicates the input signal SinUL input to the constant current supply unit **223uB** from the control device **3**. “So0” in Table 2 indicates the output signal So0 that the comparator **223a** arranged in the constant current supply unit **223uB** outputs. “Soa” in Table 2 indicates the output signal Soa that the comparator **226a** arranged in the signal level determination unit **226uB** outputs. “Sob” in Table 2 indicates the output signal Sob that the comparator **226b** arranged in the signal level determination unit **226uB** outputs. “Sod” in Table 2 indicates the output signal Sod that the comparator **226d** arranged in the signal level determination unit **226uB** outputs. “Soe” in Table 2 indicates the output signal Soe that the comparator **226e** arranged in the signal level determination unit **226uB** outputs.

(150) “Scad” in Table 2 indicates the operation result signal Scad that the AND gate **224d** arranged in the drive capability switching unit **224uB** outputs. “Scaa” in Table 2 indicates the operation result signal Scaa that the NAND gate **224a** arranged in the drive capability switching unit **224uB** outputs. “Scab” in Table 2 indicates the operation result signal Scab that the NAND gate **224b** arranged in the drive capability switching unit **224uB** outputs. “Scac” in Table 2 indicates the operation result signal Scac that the NAND gate **224c** arranged in the drive capability switching unit **224uB** outputs. “Qf” in Table 2 indicates the transistor **223f** arranged in the constant current supply unit **223uB**. “Qg” in Table 2 indicates the transistor **223g** arranged in the constant current supply unit **223uB**. “Qh” in Table 2 indicates the transistor **223h** arranged in the constant current supply unit **223uB**. “L” in Table 2 indicates that the signal level is the low level. “H” in Table 2 indicates that the signal level is the high level. “ON” in Table 2 indicates that the transistor is in the ON state. “OFF” in Table 2 indicates that the transistor is in the OFF state.

(151) TABLE-US-00002

TABLE 2	SinUL	So0	Soa	Sob	Sod	Soe	Scad	Scaa	Scab	Scac	Qf	Qg	Qh
Ssw4 > Scoc	H	L	H	T	H	L	H	H	H	H	OFF	OFF	OFF
L	H	H	L	H	L	L	L	H	H	ON	OFF	OFF	Scof <
Ssw4 < Scoc	H	L	L	H	H	L	H	H	H	OFF	OFF	OFF	L
H	L	H	H	L	H	H	L	H	H	L	H	H	L
H	OFF	ON	OFF	Ssw4 <	Scof	H	L	L	H	L	H	H	H
H	OFF	OFF	OFF	I	H	L	H	L	H	H	H	L	OFF
OFF	OFF	OFF	I	H	L	H	L	H	H	H	L	OFF	OFF
ON													

(Case where Ssw4>Scoc)

(152) First, operation of the gate drive circuit **22B** in the case where the Zener voltage of the Zener diode **6uL** is higher than the voltage generated by the voltage generation unit **226c** arranged in the signal level determination unit **226uB**, that is, in the case where the signal level of the switching signal Ssw4 is higher than the signal level of the comparison signal Scoc, will be described.

(153) When the input signal SinUL the signal level of which is the high level is input from the

control device **3** to the constant current supply unit **223uB**, the signal level of the output signal **So0** output from the comparator **223a** becomes the low level, as shown in the upper row in the row “**Ssw4>Scoc**” in Table 2.

(154) When the output signal **So0** the signal level of which is the low level is input to the other input terminals of the NAND gates **224a**, **224b**, and **224c** arranged in the drive capability switching unit **224uB**, the NAND gates **224a**, **224b**, and **224c** outputs, regardless of the signal levels of signals input to one input terminals thereof, the operation result signals **Scaa**, **Scab**, and **Scac** the signal levels of which are the high level to the gates **G** of the transistors **223f**, **223g**, and **223h** arranged in the constant current supply unit **223uB**, respectively. Thus, as shown in the upper row in the row “**Ssw4>Scoc**” in Table 2, all the transistors **223f**, **223g**, and **223h** are controlled to the OFF state. Thus, when the IGBT **211a** arranged in the semiconductor element **21u** is in the OFF state, the constant current supply unit **223uB** drives the IGBT **211a** to maintain the OFF state. In contrast, when the IGBT **211a** is in the ON state, the constant current supply unit **223uB** drives the IGBT **211a** to switch from the ON state to the OFF state. That is, the constant current supply unit **223uB** drives the IGBT **211a** to turn off.

(155) When the input signal **SinUL** the signal level of which is the low level is input from the control device **3** to the constant current supply unit **223uB**, the signal level of the output signal **So0** output from the comparator **223a** becomes the high level, as shown in the lower row in the row “**Ssw4>Scoc**” in Table 2.

(156) Since the signal level of the switching signal **Ssw4** is higher than the signal level of the comparison signal **Scoc**, the comparator **226a** outputs the output signal **Soa** the signal level of which is the high level, as shown in the lower row in the row “**Ssw4>Scoc**” in Table 2. When the output signal **Soa** the signal level of which is the high level is input to one input terminal of the NAND gate **224a** and the output signal **So0** the signal level of which is the high level is input to the other input terminal of the NAND gate **224a**, the NAND gate **224a** outputs the operation result signal **Scaa** the signal level of which is the low level to the gate **G** of the transistor **223f**, as shown in the lower row in the row “**Ssw4>Scoc**” in Table 2.

(157) In addition, as shown in the lower row in the row “**Ssw4>Scoc**” in Table 2, the comparator **226b** outputs the output signal **Sob** the signal level of which is the low level. Since the signal level of the switching signal **Ssw4** is higher than the signal level of the comparison signal **Scoc**, the comparator **226d** outputs the output signal **Sod** the signal level of which is the high level. By way of this, the output signal **Sob** the signal level of which is the low level is input to one input terminal of the AND gate **224d** and the output signal **Sod** the signal level of which is the high level is input to the other input terminal of the AND gate **224d**. Thus, as shown in the lower row in the row “**Ssw4>Scoc**”, the AND gate **224d** outputs the operation result signal **Scad** the signal level of which is the low level.

(158) When the operation result signal **Scad** the signal level of which is the low level is input to one input terminal of the NAND gate **224b** and the output signal **So0** the signal level of which is the high level is input to the other input terminal of the NAND gate **224b**, the NAND gate **224b** outputs the operation result signal **Scab** the signal level of which is the high level to the gate **G** of the transistor **223g**, as shown in the lower row in the row “**Ssw4>Scoc**” in Table 2.

(159) Further, as shown in the lower row in the row “**Ssw4>Scoc**” in Table 2, the comparator **226e** outputs the output signal **Soe** the signal level of which is the low level. When the output signal **Soe** the signal level of which is the low level is input to one input terminal of the NAND gate **224c** and the output signal **So0** the signal level of which is the high level is input to the other input terminal of the NAND gate **224c**, the NAND gate **224c** outputs the operation result signal **Scac** the signal level of which is the high level to the gate **G** of the transistor **223h**, as shown in the lower row in the row “**Ssw4>Scoc**” in Table 2.

(160) As a result, as shown in the lower row in the row “**Ssw4>Scoc**” in Table 2, the transistor **223f** is controlled to the ON state, the transistor **223g** is controlled to the OFF state, and the transistor

223h is controlled to the OFF state. Thus, when the IGBT **211a** arranged in the semiconductor element **21u** is in the OFF state, the constant current supply unit **223uB** supplies the IGBT **211a** with constant current from the transistor **223f** that has the largest current amount of constant current among the transistors **223f**, **223g**, and **223h**. By way of this, the constant current supply unit **223uB** causes the IGBT **211a** to switch to the ON state with high drive capability.

(161) In contrast, when the IGBT **211a** is in the ON state, the constant current supply unit **223uB** drives the IGBT **211a** to maintain the ON state by supplying constant current from the transistor **223f** to the IGBT **211a**.

(162) (Case where $S_{cof} < S_{sw4} < S_{coc}$)

(163) Next, operation of the gate drive circuit **22B** in the case where the Zener voltage of the Zener diode **6uL** is lower than the voltage generated by the voltage generation unit **226c** arranged in the signal level determination unit **226uB** and higher than the voltage generated by the voltage generation unit **226f** arranged in the signal level determination unit **226uB**, that is, in the case where the signal level of the switching signal S_{sw4} is lower than the signal level of the comparison signal S_{coc} and higher than the signal level of the comparison signal S_{cof} , will be described.

(164) When the input signal S_{inUL} the signal level of which is the high level is input from the control device **3** to the constant current supply unit **223uB**, the signal level of the output signal S_{o0} output from the comparator **223a** becomes the low level, as shown in the upper row in the row " $S_{cof} < S_{sw4} < S_{coc}$ " in Table 2.

(165) When the output signal S_{o0} the signal level of which is the low level is input to the other input terminals of the NAND gates **224a**, **224b**, and **224c**, the NAND gates **224a**, **224b**, and **224c** outputs, regardless of the signal levels of signals input to one input terminals thereof, the operation result signals S_{caa} , S_{cab} , and S_{cac} the signal levels of which are the high level to the gates **G** of the transistors **223f**, **223g**, and **223h**, respectively. Thus, as shown in the upper row in the row " $S_{cof} < S_{sw4} < S_{coc}$ " in Table 2, all the transistors **223f**, **223g**, and **223h** are controlled to the OFF state. Thus, when the IGBT **211a** arranged in the semiconductor element **21u** is in the OFF state, the constant current supply unit **223uB** drives the IGBT **211a** to maintain the OFF state. In contrast, when the IGBT **211a** is in the ON state, the constant current supply unit **223uB** drives the IGBT **211a** to switch from the ON state to the OFF state. That is, the constant current supply unit **223uB** drives the IGBT **211a** to turn off.

(166) When the input signal S_{inUL} the signal level of which is the low level is input from the control device **3** to the constant current supply unit **223uB**, the signal level of the output signal S_{o0} output from the comparator **223a** becomes the high level, as shown in the lower row in the row " $S_{cof} < S_{sw4} < S_{coc}$ " in Table 2.

(167) Since the signal level of the switching signal S_{sw4} is lower than the signal level of the comparison signal S_{coc} , the comparator **226a** outputs the output signal S_{oa} the signal level of which is the low level, as shown in the lower row in the row " $S_{cof} < S_{sw4} < S_{coc}$ " in Table 2. When the output signal S_{oa} the signal level of which is the low level is input to one input terminal of the NAND gate **224a** and the output signal S_{o0} the signal level of which is the high level is input to the other input terminal of the NAND gate **224a**, the NAND gate **224a** outputs the operation result signal S_{caa} the signal level of which is the high level to the gate **G** of the transistor **223f**, as shown in the lower row in the row " $S_{cof} < S_{sw4} < S_{coc}$ " in Table 2.

(168) In addition, as shown in the lower row in the row " $S_{cof} < S_{sw4} < S_{coc}$ " in Table 2, the comparator **226b** outputs the output signal S_{ob} the signal level of which is the high level. Since the signal level of the switching signal S_{sw4} is higher than the signal level of the comparison signal S_{cof} , the comparator **226d** outputs the output signal S_{od} the signal level of which is the high level. By way of this, the output signal S_{ob} the signal level of which is the high level is input to one input terminal of the AND gate **224d** and the output signal S_{od} the signal level of which is the high level is input to the other input terminal of the AND gate **224d**. Thus, as shown in the lower row in the row " $S_{cof} < S_{sw4} < S_{coc}$ ", the AND gate **224d** outputs the operation result signal S_{cad} the signal

level of which is the high level.

(169) When the operation result signal Scad the signal level of which is the high level is input to one input terminal of the NAND gate **224b** and the output signal So0 the signal level of which is the high level is input to the other input terminal of the NAND gate **224b**, the NAND gate **224b** outputs the operation result signal Scab the signal level of which is the low level to the gate G of the transistor **223g**, as shown in the lower row in the row “Scof<Ssw4<Scoc” in Table 2.

(170) In addition, as shown in the lower row in the row “Scof<Ssw4<Scoc” in Table 2, the comparator **226e** outputs the output signal Soe the signal level of which is the low level. When the output signal Soe the signal level of which is the low level is input to one input terminal of the NAND gate **224c** and the output signal So0 the signal level of which is the high level is input to the other input terminal of the NAND gate **224c**, the NAND gate **224c** outputs the operation result signal Scac the signal level of which is the high level to the gate G of the transistor **223h**, as shown in the lower row in the row “Scof<Ssw4<Scoc” in Table 2.

(171) As a result, as shown in the lower row in the row “Scof<Ssw4<Scoc” in Table 2, the transistor **223f** is controlled to the OFF state, the transistor **223g** is controlled to the ON state, and the transistor **223h** is controlled to the OFF state. Thus, when the IGBT **211a** arranged in the semiconductor element **21u** is in the OFF state, the constant current supply unit **223uB** supplies the IGBT **211a** with constant current from the transistor **223g** that has a second-largest current amount of constant current among the transistors **223f**, **223g**, and **223h**. By way of this, the constant current supply unit **223uB** causes the IGBT **211a** to switch to the ON state with the second-highest drive capability.

(172) In contrast, when the IGBT **211a** is in the ON state, the constant current supply unit **223uB** drives the IGBT **211a** to maintain the ON state by supplying constant current from the transistor **223g** to the IGBT **211a**.

(173) (Case where $Ssw4 < Scof$)

(174) Next, operation of the gate drive circuit **22B** in the case where the Zener voltage of the Zener diode **6uL** is lower than the voltages generated by the respective ones of the voltage generation units **226c** and **226f** arranged in the signal level determination unit **226uB**, that is, in the case where the signal level of the switching signal Ssw4 is lower than the signal levels of the respective ones of the comparison signals Scoc and Scof, will be described.

(175) When the input signal SinUL the signal level of which is the high level is input from the control device **3** to the constant current supply unit **223uB**, the signal level of the output signal So0 output from the comparator **223a** becomes the low level, as shown in the upper row in the row “Ssw4<Scof” in Table 2.

(176) When the output signal So0 the signal level of which is the low level is input to the other input terminals of the NAND gates **224a**, **224b**, and **224c**, the NAND gates **224a**, **224b**, and **224c** outputs, regardless of the signal levels of signals input to one input terminals thereof, the operation result signals Scaa, Scab, and Scac the signal levels of which are the high level to the gates G of the transistors **223f**, **223g**, and **223h**, respectively. Thus, as shown in the upper row in the row “Ssw4<Scof” in Table 2, all the transistors **223f**, **223g**, and **223h** are controlled to the OFF state. Thus, when the IGBT **211a** arranged in the semiconductor element **21u** is in the OFF state, the constant current supply unit **223uB** drives the IGBT **211a** to maintain the OFF state. In contrast, when the IGBT **211a** is in the ON state, the constant current supply unit **223uB** drives the IGBT **211a** to switch from the ON state to the OFF state. That is, the constant current supply unit **223uB** drives the IGBT **211a** to turn off.

(177) When the input signal SinUL the signal level of which is the low level is input from the control device **3** to the constant current supply unit **223uB**, the signal level of the output signal So0 output from the comparator **223a** becomes the high level, as shown in the lower row in the row “Ssw4<Scof” in Table 2.

(178) Since the signal level of the switching signal Ssw4 is lower than the signal level of the

comparison signal **Scoc**, the comparator **226a** outputs the output signal **Soa** the signal level of which is the low level, as shown in the lower row in the row “**Ssw4<Scof**” in Table 2. When the output signal **Soa** the signal level of which is the low level is input to one input terminal of the NAND gate **224a** and the output signal **So0** the signal level of which is the high level is input to the other input terminal of the NAND gate **224a**, the NAND gate **224a** outputs the operation result signal **Scaa** the signal level of which is the high level to the gate G of the transistor **223f**, as shown in the lower row in the row “**Ssw4<Scof**” in Table 2.

(179) In addition, as shown in the lower row in the row “**Ssw4<Scof**” in Table 2, the comparator **226b** outputs the output signal **Sob** the signal level of which is the high level. Since the signal level of the switching signal **Ssw4** is lower than the signal level of the comparison signal **Scof**, the comparator **226d** outputs the output signal **Sod** the signal level of which is the low level. By way of this, the output signal **Sob** the signal level of which is the high level is input to one input terminal of the AND gate **224d** and the output signal **Sod** the signal level of which is the low level is input to the other input terminal of the AND gate **224d**. Thus, as shown in the lower row in the row “**Ssw4<Scof**”, the AND gate **224d** outputs the operation result signal **Scad** the signal level of which is the low level.

(180) When the operation result signal **Scad** the signal level of which is the low level is input to one input terminal of the NAND gate **224b** and the output signal **So0** the signal level of which is the high level is input to the other input terminal of the NAND gate **224b**, the NAND gate **224b** outputs the operation result signal **Scab** the signal level of which is the high level to the gate G of the transistor **223g**, as shown in the lower row in the row “**Ssw4<Scof**” in Table 2.

(181) Since the signal level of the switching signal **Ssw4** is lower than the signal level of the comparison signal **Scof**, the comparator **226e** outputs the output signal **Soe** the signal level of which is the high level, as shown in the lower row in the row “**Ssw4<Scof**” in Table 2. When the output signal **Soe** the signal level of which is the high level is input to one input terminal of the NAND gate **224c** and the output signal **So0** the signal level of which is the high level is input to the other input terminal of the NAND gate **224c**, the NAND gate **224c** outputs the operation result signal **Scac** the signal level of which is the low level to the gate G of the transistor **223h**, as shown in the lower row in the row “**Ssw4<Scof**” in Table 2.

(182) As a result, as shown in the lower row in the row “**Ssw4<Scof**” in Table 2, the transistor **223f** is controlled to the OFF state, the transistor **223g** is controlled to the OFF state, and the transistor **223h** is controlled to the ON state. Thus, when the IGBT **211a** arranged in the semiconductor element **21u** is in the OFF state, the constant current supply unit **223uB** supplies the IGBT **211a** with constant current from the transistor **223h** that has the smallest current amount of constant current among the transistors **223f**, **223g**, and **223h**. By way of this configuration, the constant current supply unit **223uB** causes the IGBT **211a** to switch to the ON state with the lowest drive capability.

(183) In contrast, when the IGBT **211a** is in the ON state, the constant current supply unit **223uB** drives the IGBT **211a** to maintain the ON state by supplying constant current from the transistor **223h** to the IGBT **211a**.

(184) As described above, the gate drive circuit **22B** and the intelligent power module **2B** are capable of changing drive capability to drive the IGBTs (corresponding to the IGBTs **201a** and **211a**) arranged in the semiconductor elements corresponding to the U-phase, the V-phase, and the W-phase of each of the upper arm and the lower arm only by connecting the Zener diodes (corresponding to the Zener diodes **6uH** to **6wL** in the above-described first embodiment) having predetermined Zener voltages to the switching signal input terminals (corresponding to the switching signal input terminals **Tsw1** to **Tsw6** in the above-described first embodiment) corresponding to the U-phase, the V-phase, and the W-phase of each of the upper arm and the lower arm.

(185) As described in the foregoing, since the gate drive circuit **22B** as the drive circuit of

switching elements according to the present embodiment has a similar configuration to that of the gate drive circuit **22A** according to the above-described first embodiment, the gate drive circuit **22B** is capable of selecting a drive capability to drive the switching elements in a simple method. (186) The intelligent power module **2B** according to the present embodiment includes the IGBTs **201a** and **211a** and the gate drive circuit **22B**.

(187) By way of this, the intelligent power module **2B** is capable of selecting a drive capability to drive the switching elements in a simple method.

(188) (Variation)

(189) Although, in the present embodiment, in each of the constant current supply units **221uB** to **223wB** arranged in the gate drive circuit **22B**, current amounts of constant currents supplied to an IGBT **211a** from the transistors **223f**, **223g** and **223h** are different from each other, the current amounts of the constant currents may be the same. In this case, when the signal level of the output signal **Soa** input from the comparator **226a** arranged in the signal level determination unit **226uB** is the high level, the signal level of the operation result signal **Scad** input from the AND gate **224d** is the low level, and the signal level of the output signal **Soe** is the low level, each of the drive capability switching units **222uB** to **224wB** controls all the transistors **223f**, **223g**, and **223h** to the ON state. In addition, when the signal levels of the output signals **Soa** and **Soe** input from the comparators **226a** and **226e** are the low level and the signal level of the operation result signal **Scad** is the high level, each of the drive capability switching units **222uB** to **224wB** controls any two of the transistors **223f**, **223g**, and **223h** to the ON state and controls the remaining one to the OFF state. Further, when the signal level of the output signal **Soa** input from the comparator **226a** is the low level, the signal level of the operation result signal **Scad** is the low level, and the signal level of the output signal **Soe** input from the comparator **226e** is the high level, each of the drive capability switching units **222uB** to **224wB** controls the transistor **223h** to the ON state and controls the transistors **223f** and **223g** to the OFF state.

(190) As the number of transistors in the ON state among the transistors **223f**, **223g**, and **223h** increases, each of the constant current supply units **221uA** to **223wB** is capable of supply the IGBTs **201a** and **211a** with constant currents with larger current amounts. By way of this, the gate drive circuit **22B** and the intelligent power module **2B** are capable of changing drive capability to drive the IGBTs **201a** and **211a** only by connecting the Zener diodes **6uH** to **6wL** having predetermined Zener voltages to the switching signal input terminals **Tsw1** to **Tsw6**, respectively.

(191) The present invention is not limited to the above-described embodiments, and various modification are possible.

(192) Although, in the above-described first embodiment and the above-described second embodiment, the signal level of each switching signal is set by connecting a Zener diode to a corresponding switching signal input terminal, the present invention is not limited to the configuration. For example, a connection point between a plurality of resistance elements connected in series between the power source terminal and the reference potential terminal may be connected to the switching signal input terminal. In this case, since voltage based on resistance division by the plurality of resistance elements is input to the switching signal input terminal, it is possible to set the signal level of the switching signal.

(193) Although, in the above-described first embodiment and the above-described second embodiment, it is configured such that at least one of a plurality of transistors that are arranged in each constant current supply unit and supply constant currents having current amounts different from one another is controlled to the ON state, the present invention is not limited to the configuration. For example, even when a plurality of transistors supplying constant currents having current amounts different from one another coexist, one or two or more transistors may be appropriately controlled to the ON state, as in the variations of the above-described first embodiment and the above-described second embodiment. This configuration allows the drive capability of the gate drive circuit to be changed to more levels than the number of transistors

arranged in each constant current supply unit.

(194) Although, in the above-described first embodiment and the above-described second embodiment, the drive capability of the gate drive circuit is switched by a plurality of transistors, the present invention is not limited to the configuration. For example, one transistor may be arranged in each constant current supply unit and, by changing the voltage level input to the gate of the transistor according to the signal level of a switching signal, the current amount of current supplied from the transistor may be switched. In this case, the drive capability of the gate drive circuit can also be switched.

(195) Although the intelligent power modules 2A and 2B according to the above-described first embodiment and the above-described second embodiment include IGBTs as switching elements, the present invention is not limited to the configuration. The intelligent power modules 2A and 2B may include, in place of IGBTs, either bipolar transistors or power MOS transistors as switching elements.

(196) The technical scope of the present invention is not limited to the illustrated and described exemplary embodiments, but includes all embodiments causing an effect equivalent to the object of the present invention.

(197) Furthermore, the technical scope of the present invention is not limited to combinations of features of the inventions defined by the claims, but can be defined by all desired combinations of specific features among all the disclosed features.

REFERENCE SIGNS LIST

(198) 1A Power conversion device 2A, 2B Intelligent power module 2U U-phase output arm 2V V-phase output arm 2W W-phase output arm 3 Control device 4 AC power unit 5 Smoothing capacitor 6uH, 6vH, 6wH, 6uL, 6vL, 6wL Zener diode 7 Motor 8L Power source 20v, 20w, 21u, 21v, 21w Semiconductor element 22A, 22B Gate drive circuit 201, 211 IGBT chip 201a, 211a IGBT 202, 212 Freewheel diode 211b Temperature detection element 221uA, 221uB, 221vA, 221vB, 221wA, 221wB Constant current supply unit 222uA, 222uB, 222vA, 222vB, 222wA, 222wB Drive capability switching unit 223a, 225e-1, 225f-1, 225g-1, 226a, 226b, 226d, 226e Comparator 223b, 225e-2, 225f-2, 225g-2, 226c, 226f Voltage generation unit 223c, 223d, 225g-3, 225h Resistance element 223e, 225c Constant current source 223f, 223g, 223h, 225b, 223h Transistor 223uA, 223uB, 223vA, 223vB, 223wA, 223wB Constant current supply unit 224a, 224b, 224c NAND gate 224d AND gate 224uA, 224uB, 224vA, 224vB, 224wA, 224wB Drive capability switching unit 225, 225uA, 225vA, 225wA Protection unit 225a Temperature detection unit 225d Alarm signal generation circuit 225e Control voltage detection circuit 225f Temperature detection circuit 225g Current detection circuit 226A, 226B Signal level determination unit 211b Temperature detection element ALM, ALMu, ALMv, ALMw Alarm signal Ln Negative electrode-side line Lp Positive electrode-side line Scaa, Scab, Scac, Scad Operation result signal Sco, Scoc, Scof Comparison signal SinUH, SinUL, SinVH, SinVL, SinWH, SinWL Input signal So0, Soa, Sob, Sod, Soe Output signal Ssw1, Ssw2, Ssw3, Ssw4, Ssw5, Ssw6 Switching signal T22, T223 Reference potential terminal Tiuh, Tiul, Tivh, Tivl, Tiwh, Tiwl Signal input terminal Tn Negative electrode-side power input terminal Tp Positive electrode-side power input terminal Tsw1, Tsw2, Tsw3, Tsw4, Tsw5, Tsw6 Switching signal input terminal TU U-phase output terminal TV V-phase output terminal TvIn Power source terminal TW W-phase output terminal V0 Comparison voltage VFO, VFOu, VFOv, VFOw Alarm signal output terminal

Claims

1. A drive circuit for a switching element comprising: a constant current supply to receive an input signal from outside of the drive circuit and to supply constant current to a control signal input terminal of the switching element according to a signal level of the input signal; a switching signal input terminal to receive a switching signal; a signal level determiner, in communication with the

switching signal input terminal, to determine a signal level of the switching signal by comparing the switching signal with a comparison signal; and a drive capability switch, in communication with the signal level determiner and the constant current supply, to switch drive capability of the constant current supply by changing a current amount of the constant current supplied by the constant current supply based on a determination result in the signal level determiner and the signal level of the input signal, wherein the switching element switches to an ON state according to the current amount of the constant current supplied to the switching element by the constant current supply.

2. The drive circuit for the switching element according to claim 1, wherein the constant current supply includes a plurality of transistors configured to supply the control signal input terminal with the constant current.

3. The drive circuit for the switching element according to claim 2, wherein the drive capability switch controls, based on the determination result and a signal level of the input signal, at least one of the plurality of transistors to the ON state.

4. The drive circuit for the switching element according to claim 3, wherein the signal level determiner includes a plurality of comparators configured to compare a comparison signal, the comparison signal being set to a predetermined signal level, with the switching signal, and at least two comparators among the plurality of comparators to output output signals having signal levels inverted to each other to the drive capability switch.

5. The drive circuit for the switching element according to claim 4, wherein the drive capability switch includes a first operation unit configured to perform a logical operation on a signal level of the output signal of one of the two comparators and a signal level of a signal based on the input signal and a second operation unit configured to perform a logical operation on a signal level of the output signal of the other of the two comparators or a signal level of a signal based on the output signal and the signal level of a signal based on the input signal, the first operation unit outputs an operation result signal to a control signal input terminal of one of the two transistors, and the second operation unit outputs an operation result signal to a control signal input terminal of the other of the two transistors.

6. The drive circuit for the switching element according to claim 1 comprising: a temperature detection unit configured to detect temperature of the switching element; a current detection unit configured to detect current flowing through the switching element; a control voltage detection unit configured to detect control voltage to control the constant current supply; and an alarm signal generation unit configured to generate an alarm signal notifying that at least one of the temperature detection unit, the current detection unit, and the control voltage detection unit detected an abnormality.

7. An intelligent power module comprising: the drive circuit for the switching element according to claim 6; and the switching element.

8. An intelligent power module comprising: the drive circuit for the switching element according to claim 1; and the switching element.
